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The Intrinsic Equilibrium Demographic Distribution of Galactic Pulsars: Steady-State Continuity Solutions, Active Inventory, and Discovery Forecasts

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ABSTRACT

Forward-evolving the 2D continuity operator under dipole braking, crustal dissipation, and a continuous death-valley sink, we determine the intrinsic equilibrium demographic distribution $n_{\rm true}(P, B)$ of Galactic radio pulsars. Although births peak at tens of milliseconds ($P_0 \approx 48$ ms, median 59 ms), decelerating torque advection ($v_x \propto P^{-(n-1)}$) causes pulsars to accumulate at second-scale periods: residency timescales $\tau_{\rm res} \propto P^2/B^2$ rise by $\sim 10^5$ from birth (~ 270 yr/dex) to maturity (~ 23 Myr/dex). The intrinsic population peaks at $P = 1.61$ s with median 1.38 s (propagated bootstrap mean 1.26 ± 0.05 s; 10–90% interval $[0.45, 3.97]$ s), spending $\approx 71\%$ of active life in $[0.5, 3.0]$ s with a 17% ultra-slow tail ($9.8\% \pm 1.7\%$ bootstrap). Across pair thresholds, the active inventory spans $[0.4, 1.7] \times 10^5$ pulsars (fiducial $(0.82 \pm 0.28) \times 10^5$, compounding supernova rate and bootstrap source errors), with $(1.13 \pm 0.38) \times 10^8$ dead radio corpses. We evaluate this equilibrium across four in-sample consistency checks: literature inventories, cylinder currents, vertical kinematic drift ($W_1 = 0.031$ dex), and counterfactual slow injection. Applying calibrated selection reproduces the observed ATNF median (0.62 s), while an inverse-volume-weighted ATNF median independently shifts to 1.01 s. Next-generation surveys will probe this slow reservoir: FAST and SKA1-Mid discovery medians shift toward 0.83 s and 1.03 s, with SKA1-Mid reaching an ideal geometric ceiling of $\sim$ 8,100 pulsars ($\sim 10\%$ of active stars; realistic $\sim$ 6,600–7,000), while $\approx 90\%$ remain undetectable from beaming and sky limits.

Keywords: Astrostatistics (1882) — Pulsars (1306) — Neutron stars (1108) — Galaxy structure (622) — Stellar evolution (1599)

1. INTRODUCTION

The statistical demographics of isolated Galactic radio pulsars are governed by two distinct physical distributions: the *birth distribution* $S(P_0, B_0)$, which specifies the injection rate of newborn neutron stars into phase space per unit spin period and surface magnetic field, and the *intrinsic equilibrium demographic distribution* $n_{\rm true}(P, B)$, which describes the actual, physical distribution of active pulsars residing in the Milky Way at the present epoch.

In population synthesis literature, these two concepts are frequently blurred. Forward Monte Carlo studies typically assume a parametric functional form for the birth period distribution—such as a broad unimodal Gaussian centered at $\langle P_0 \rangle \approx 0.30$ s (Faucher-Giguère & Kaspi 2006) or a log-normal distribution (Lorimer et al. 2006)—and adjust initial parameters until the selection-filtered mock sample matches the one-dimensional (1D) period histogram of observed surveys like the Parkes Multibeam Pulsar Survey (PMPS; Manchester et al. 2005). Conversely, historical studies examining the 1D *pulsar current* $J(P) \equiv \dot{P}\, n(P)$ (Phinney & Blandford 1981; Narayan & Vivekanand 1982) argued that the observed prominence of pulsars with $P \sim 0.5$–1.0 s required that 40%–75% of all pulsars are "injected" into the population at slow spin periods, either born directly with $P_0 \gtrsim 0.5$ s or emerging into radio visibility only after a delay of $\tau \sim 10^5$–10^6 yr.

Recently, Bindra (2026a) (hereafter Paper I) developed a weak-form, non-parametric two-dimensional (2D) continuity inversion framework that maps the observed distribution of the Australia Telescope National Facility (ATNF) Pulsar Catalogue back to the initial injection plane. That analysis demonstrated that the empirical catalogue is consistent with a predominantly fast-birth scenario: the reconstructed birth distribution peaks sharply at $P_0 \approx 48$ ms (with an intrinsic baseline tail of 10.8%–11.1% born above 0.5 s, leaving an allowable incremental headroom of $\Delta f_{\rm slow} \leq 11.2\%$ within the total 95% upper bound of $f_{\rm slow}^{\rm birth} \leq 22\%$), disfavoring the classical $\geq 40\%$ injection hypothesis.

However, establishing that pulsars are born fast raises an immediate, fundamental astrophysical question: *If*

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neutron stars enter the Galaxy spinning at tens of milliseconds, what is the actual, physical distribution of pulsars in the Milky Way today? In any flux-limited catalogue, the observed distribution $n_{\rm obs}(P, B)$ is heavily distorted by selection effects: radio luminosity scales with spin-down power ($\dot{E} \propto B^2 P^{-4}$), beam opening angles shrink for slower rotators ($f_{\rm beam} \propto P^{-1/2}$; Tauris & Manchester 1998), and radiometer sensitivity degrades at short periods due to interstellar dispersion and multipath pulse broadening. Consequently, the canonical isolated ATNF sample ($N = 2,356$) represents only a tiny, biased fraction (2.9% of the fiducial $N_{\rm Gal}^{\rm active} = 0.82 \times 10^5$ population for $f_{\rm radio} = 0.75$, or 2.2% in the pure neutron-star limit 1.09×10^5) of the active Galactic population. Model-independently, assigning each empirical ATNF pulsar an inverse-volume and beaming weight $w_i \propto 1/[V_{\rm max}(L_i) f_{\rm beam}(P_i)]$ shifts the empirical median from 0.62 s directly to 1.01 s (with 50% of weighted stars having $P > 1$ s), demonstrating from raw data alone that the underlying Galactic population is slow-dominated.

Moreover, to assess how this equilibrium state behaves physically and distinguish internal mathematical consistency from external physical behavior, we draw a clear methodological distinction between *internal closure* and *forward consistency checks*. Convolving our forward-evolved demographic equilibrium with the forward selection filter serves as an internal closure test: it confirms that the transport-inversion mathematical loop closes self-consistently by recovering the empirical ATNF catalogue ($W_1 = 0.037$ dex on $\log_{10} P$). To assess whether the equilibrium distribution represents physical reality rather than an artifact of model calibration, the framework is evaluated across four forward in-sample consistency checks and demographic explorations—in-sample vertical kinematic scale-height migration into the halo ($|b| > 5°$), volume-limited 1D currents in local cylinders ($d \leq 1.5$ kpc), literature inventory consistency, and comparative demographic sensitivity to alternative injection scenarios. Because underlying parameters governing crustal field decay ($h_0, B_{\rm floor}$) and empirical survey completeness were calibrated on the ATNF catalogue in Paper I, these checks are not fully blind from the underlying observational data; rather, they serve as consistency checks verifying that the 2D steady-state continuity solution exhibits physically sensible behavior across spatial, kinematic, and demographic projections.

In this work, we present a complete forward continuity analysis that determines the intrinsic equilibrium distribution $n_{\rm true}(P, B)$ of Galactic radio pulsars. While the 1D scaling $n(P) \propto P$ under decelerating dipole torques has long been recognized as an elementary property of rotational kinematics (Phinney & Blandford 1981), self-consistently determining the true 2D phase-space distribution, total inventory, and survey discovery forecasts directly from an empirical fast-birth distribution requires solving the continuous transport problem across magnetic field decay, core-crustal physics, and pair-production death boundaries (Tauris & Konar 2001; Konar 2017). This framework provides a direct continuum counterpart to forward 3D Monte Carlo synthesis codes (Faucher-Giguère & Kaspi 2006; Popov et al. 2010; Gullón et al. 2014; Cieślar et al. 2020). Where forward Monte Carlo studies simulate discrete stochastic realizations under assumed parametric birth functions—for instance, Popov et al. (2010) incorporate magneto-thermal evolution and infer an intrinsic population peaking at $P \sim 1$–1.5 s from parametric log-normal births, while Gullón et al. (2014) and Cieślar et al. (2020) explore the interplay between thermal cooling and crustal field decay—our 2D continuum approach solves the transport operator deterministically. By evolving an empirically reconstructed, non-parametric fast-birth distribution through high-resolution Eulerian marching and continuous-time Markov-renewal Lagrangian simulations over $T = 10^8$ yr, we show how decelerating torque advection naturally produces an equilibrium population whose median spin period is $P = 1.38$ s from a fast-birth peak ($P_0 \approx 48$ ms) without Monte Carlo Poisson noise or ad hoc slow injection.

The structure of this paper is as follows. In Section 2, we formulate the 2D continuity operator, characteristic drift velocities, crustal magnetic floors, and boundary sinks. In Section 3, we compute the intrinsic demographic distribution $n_{\rm true}(P, B)$, derive the per-decade residency timescale $\tau_{\rm res}(P)$, explore parameter sensitivities (braking indices n, sink widths, field decay rates, and slow-birth inflation), and calculate total Galactic energy budgets and graveyard populations. In Section 4, we evaluate the equilibrium population against internal closure and physical cross-checks: total inventory normalization, 1D steady-state currents in volume-limited cylinders, out-of-sample high-latitude kinematic drift, and counterfactual slow-injection sensitivity. In Section 5, we demonstrate numerical convergence between Eulerian continuum and Fortran Lagrangian realizations under matched extinction operators. In Section 6, we present discovery projections for FAST and SKA1-Mid. We conclude in Section 7.

2. MATHEMATICAL FRAMEWORK OF THE EQUILIBRIUM DISTRIBUTION

2.1. *The 2D Phase-Space Continuity Equation*

Let $n(P, B, t)$ denote the number density of active, radio-loud neutron stars per unit spin period P and surface dipole magnetic field B across the Galaxy at epoch t. Assuming that the star-formation rate in the Galactic thin disk has remained stationary over the typical active pulsar lifetime ($\tau_{\rm active} \sim 7$ Myr $\ll \tau_{\rm disk} \sim 10$ Gyr), the pulsar population satisfies the stationary continuity equation ($\partial_t n = 0$):

$$\nabla \cdot \big(\mathbf{v}(P, B)\, n_{\rm true}(P, B)\big) + \Lambda(P, B)\, n_{\rm true}(P, B) = S(P, B), \tag{1}$$

where $\mathbf{v}(P,B) = (\dot{P}, \dot{B})$ is the deterministic phase-space trajectory velocity, $\Lambda(P,B) = \kappa + d_{\rm sink}(P,B)$ encapsulates particle sinks (kinematic disk escape κ and pair-production shutdown $d_{\rm sink}$), and $S(P,B)$ is the birth injection rate per unit phase space.

Transforming into logarithmic coordinates $x \equiv \log_{10}(P/{\rm s})$ and $y \equiv \log_{10}(B/{\rm G})$, the continuity equation preserves probability flux via $f(x,y)\,dx\,dy = n(P,B)\,dP\,dB$, leading to:

$$\frac{\partial}{\partial x}\big[v_x(x,y)\,f(x,y)\big] + \frac{\partial}{\partial y}\big[v_y(x,y)\,f(x,y)\big] + \Lambda(x,y)\,f(x,y) = S(x,y), \tag{2}$$

where $v_x \equiv \dot{x} = \dot{P}/(P\ln 10)$ and $v_y \equiv \dot{y} = \dot{B}/(B\ln 10)$.

2.2. *Characteristic Flow Velocities: Torques and Crustal Magnetic Floors*

Rotational energy loss due to magnetospheric dipole radiation and relativistic particle winds gives the generalized spin-down law parameterized by braking index $n \equiv \Omega\ddot{\Omega}/\dot{\Omega}^2$:

$$\dot{P} = \frac{8\pi^2 R^6 B^2}{3Ic^3 P}\left(\frac{P}{P_{\rm ref}}\right)^{3-n} \equiv \frac{B^2}{K_{\rm spin} P}\left(\frac{P}{P_{\rm ref}}\right)^{3-n}, \tag{3}$$

where $K_{\rm spin} \equiv (3.2\times10^{19}\ {\rm G\ s^{-1/2}})^2 = 1.024\times10^{39}\ {\rm G^2\ s^{-1}}$ for a canonical neutron star with moment of inertia $I = 10^{45}\ {\rm g\ cm^2}$ and radius $R = 10$ km. For canonical magnetic dipole braking ($n = 3$), the spin-down drift velocity in logarithmic coordinates (in units of ${\rm Myr^{-1}}$) is:

$$v_x(x,y) = \frac{\mathcal{C}_{\rm Myr}\,10^{2(y-x)}}{K_{\rm spin}\ln 10} > 0, \tag{4}$$

where $\mathcal{C}_{\rm Myr} = 3.15576\times10^{13}\ {\rm s\ Myr^{-1}}$. Crucially, $v_x(x,y)$ is strictly positive across the entire physical domain: pulsars advect monotonically from left to right (increasing P). For a general braking index n, $v_x(x,y) \propto 10^{2y-(n-1)x} = B^2 P^{-(n-1)}$, demonstrating that pulsars decelerate as they spin down for any physical braking index $n > 1$.

Magnetic field evolution in neutron star crusts proceeds via Hall drift and Ohmic dissipation (Konar & Bhattacharya 1997, 1999; Pons et al. 2009; Viganò et al. 2013; Gourgouliatos & Cumming 2014). While young magnetars undergo rapid non-linear Hall cascades and thermal cooling during their first $\sim 10^5$ yr with effective dissipation timescales $\tau_{\rm Ohm} \sim 10^6$ yr, mature isolated radio pulsars cool below $\sim 10^7$ K, where electron-impurity scattering in the deep crystalline solid crust dominates electrical resistivity. For low impurity parameters ($Q_{\rm imp} \equiv n_{\rm ion}^{-1}\sum_i n_i(Z_i - \langle Z\rangle)^2 \lesssim 0.1$–$1$), microphysical models show that secular Ohmic dissipation in the deep crust proceeds on long timescales $\tau_B \sim 10^7$–10^8 yr ($\sim$ 50–100 Myr; Urpin & Muslimov 1992; Pons et al. 2009; Konar 2017), eventually asymptoting to a core-supported or impurity-limited bottom magnetic floor $B_{\rm floor} \approx 10^{11.0}$–$10^{11.3}$ G (Urpin et al. 1998; Konar 2017; Igoshev & Popov 2014). Furthermore, as shown by Tauris & Konar (2001), the coupled evolution of crustal Ohmic dissipation and torque decay governs the evolutionary tracks of isolated pulsars across the P–$\dot{P}$ plane, causing the effective braking index $n_{\rm eff} \equiv 2 - 2(P\dot{B})/(\dot{P}B)$ to deviate from $n = 3$ and modifying the residency time within observable phase-space windows. We parameterize this downward drift rate as:

$$\dot{B} = -h_0 B\left[1 - \left(\frac{B_{\rm floor}}{B}\right)\right]_+, \tag{5}$$

where $[z]_+ \equiv \max(0, z)$. In logarithmic phase space, the downward velocity is:

$$v_y(y) = -\frac{h_0}{\ln 10}\max\left[0,\ 1 - 10^{y_{\rm floor}-y}\right] \le 0. \tag{6}$$

Following the empirical calibration in Bindra (2026a) (Paper I), we adopt the canonical value $h_0 = 0.015\ {\rm Myr^{-1}}$ ($\tau_B \approx 67$ Myr), together with global secular loss $\kappa = 0.008\ {\rm Myr^{-1}}$ ($\tau_\kappa \approx 125$ Myr) and crustal floor $y_{\rm floor} = 11.20$ ($B_{\rm floor} \approx 1.6\times10^{11}$ G). We note that in Paper I, the weak-form continuity inversion was formulated under a minimal baseline ($d_{\rm crit} = 11.23, \kappa = 0.005\ {\rm Myr^{-1}}$, without a crustal B-floor). In this paper, we explore forward transport under both the exact matched inversion operator ($d_{\rm crit} = 11.23, \kappa = 0.005\ {\rm Myr^{-1}}$, no floor; Section 5) and the canonical curvature radiation model ($d_{\rm crit} = 11.45, \kappa = 0.008\ {\rm Myr^{-1}}, B_{\rm floor} = 10^{11.20}$ G), showing that the slow demographic equilibrium ($P_{\rm med} > 1.3$ s) is a universal forward outcome across both formulations. For a 1D Maxwellian kick dispersion $\sigma_{\rm 1D} = 265\ {\rm km\ s^{-1}}$ (Hobbs et al. 2005), the 3D speed distribution yields $P(v > v_{\rm esc}) \approx 23.0\%$ for an escape velocity of $v_{\rm esc} \approx 550\ {\rm km\ s^{-1}}$, meaning that $\approx 77.0\%$ of all core-collapse compact remnants remain gravitationally bound within the Galactic potential. The phenomenological secular loss rate $\kappa = 0.0080\ {\rm Myr^{-1}}$ ($\tau_{\rm esc} = 1/\kappa \approx 125$ Myr) parameterizes the slow global loss of stars from the Galactic active population due to hyperbolic kick unbinding and gradual evaporation into the distant halo over long timescales. Setting $\kappa = 0$ shifts the total active Galactic inventory $N_{\rm Gal}^{\rm active}$ by only 5.7% ($1 - e^{-\kappa\langle\tau_{\rm active}\rangle} = 1 - e^{-0.008\times7.28} \approx 0.057$), confirming that global secular unbinding is a minor perturbation on the total inventory relative to pair-production extinction. We distinguish this global unbinding rate κ from spatial vertical migration across Galactic latitudes: active pulsars that migrate away from the thin disk into the halo remain active Galactic stars. In renewal demographic equilibrium with distributed lifetimes, the stationary snapshot age distribution satisfies $p_{\rm age}(t) \propto \mathbb{P}(\tau > t)$, with mean age $\langle t\rangle =$

$\mathbb{E}[\tau^2]/(2\mathbb{E}[\tau]) \geq 6.7$ Myr across the intrinsic population (median ≈ 4.8 Myr; Section 4.3), whereas flux-limited observable pulsars are systematically younger (median ≈ 3.4 Myr). With a mean absolute vertical drift speed of $\langle|v_z|\rangle \approx 0.216$ kpc Myr^{-1} (≈ 211 km s^{-1}), this age-dependent vertical dispersion drives mature pulsars into the halo ($|z| \sim 1$–2 kpc), which we model explicitly via ballistic vertical kinematics rather than absorbing it into κ. In terms of physical timescales, the global secular escape timescale $\tau_\kappa \equiv 1/\kappa \approx 125$ Myr is nearly 18× longer than the population-averaged active lifetime ($\langle\tau_{\rm active}\rangle \approx 7.28$ Myr), confirming that secular unbinding is a minor perturbation on demographics. In phase space, local spin-down advection times across a logarithmic decade $\tau_{\rm res} = 1/v_x \sim 270$ yr–23 Myr are substantially shorter than τ_κ throughout active life, while pair-production extinction ($d_{\rm sink} \sim 5$ Myr^{-1}, $\tau_{\rm sink} \sim 0.2$ Myr) dominates the late-time boundary.

Physically, Equation 5 describes the asymptotic cessation of crustal Ohmic dissipation ($\dot{B} \to 0$ as $B \to B_{\rm floor}$), as detailed in microphysical calculations of crustal electrical conductivity and impurity parameters (Konar 2017); it represents an evolutionary floor on crustal field decay rather than an exclusionary phase-space boundary. In the ATNF catalogue, 190 classical pulsars (8.06%) have inferred dipole fields below $10^{11.20}$ G (extending down to $\sim 10^9$ G). These objects represent neutron stars born with low seed fields, core-dominated magnetic field topologies that bypass crustal Ohmic dissipation (Urpin & Muslimov 1992; Konar & Bhattacharya 1999), or mildly recycled pulsars. Once a pulsar reaches or starts below $B_{\rm floor}$, its magnetic field remains constant ($v_y = 0$), causing it to drift horizontally in period along pure spin-down trajectories $\dot{P} \propto 1/P$.

2.3. *Pair-Production Death Line Sinks and the Pulsar Death Valley*

Radio emission ceases when magnetic pair cascades above the polar cap can no longer sustain plasma discharge. Following Space-Charge-Limited Flow (SCLF) models (Chen & Ruderman 1993; Zhang et al. 2000), the nominal pair-production boundary corresponds to a threshold voltage parameterized by:

$$\Gamma(x,y) \equiv y - 2x = \log_{10}\left(\frac{B}{P^2}\right) = 11.45. \qquad (7)$$

However, theoretical and observational studies of pulsars approaching this boundary demonstrate that radio shutoff is not a knife-edge step function; rather, pulsars enter a finite "death valley" characterized by voltage starvation, high nulling fractions, and intermittent radio emission (Chen & Ruderman 1993; Zhang et al. 2000; Konar 2017; Konar & Deka 2019; Abhishek et al. 2022). In the empirical ATNF catalogue ($N = 2,356$ classical isolated pulsars), 137 pulsars (5.81%) lie below the nominal $\Gamma = 11.45$ boundary, and 51 (2.16%) lie below the classical dipole threshold $\Gamma = 11.23$, most prominently the 8.51-second slow rotator PSR J2144-3933 ($\log_{10}(B/P^2) = 10.46$; Young et al. 1999). In modern electrodynamic models, such sub-Γ radio emission is sustained by localized non-dipolar surface multipoles, highly curved magnetic field lines near the stellar surface, or intermittent spark discharge near the polar cap (Chen & Ruderman 1993; Zhang et al. 2000; Konar 2017; Abhishek et al. 2022). To represent this physical death valley and prevent artificial edge reflections, we model pair-production extinction as a continuous absorbing logistic sink with width $\Delta_{\rm width} = 0.10$ dex:

$$d_{\rm sink}(x,y) = \frac{A_{\rm sink}}{1 + \exp\left[\frac{(y-2x)-d_{\rm crit}}{\Delta_{\rm width}}\right]}, \qquad (8)$$

where $A_{\rm sink} = 5.0$ Myr^{-1} is the fiducial extinction rate amplitude, $d_{\rm crit} = 11.45$ is the nominal threshold, and $\Delta_{\rm width} = 0.10$ dex is the transition width. We caution that fitting an empirical logistic model directly to the lower edge of the observed ATNF Γ distribution conflates intrinsic radio extinction with survey flux thresholds, because spin-down power drops as $\dot{E} \propto (B/P^2)^2 = 10^{2\Gamma}$ and radio luminosity scales as $L \propto \dot{E}^{0.45}$. An empirical fit to the observed cutoff yields $d_{\rm crit} \approx 11.48 \pm 0.03$ dex and $\Delta_{\rm width} \approx 0.16 \pm 0.02$ dex, reflecting this coupled extinction-flux boundary. Taken at face value, adopting $\Delta_{\rm width} = 0.16$ dex increases pre-boundary dissipation, shifting the mean active lifetime to $\langle\tau_{\rm active}\rangle \approx 5.3$ Myr and active inventory to $N_{\rm Gal}^{\rm active} \approx 0.6 \times 10^5$, which falls comfortably within our forward systematic uncertainty envelope ($[0.53, 1.25] \times 10^5$). Furthermore, the extinction amplitude $A_{\rm sink} = 5.0$ Myr^{-1} (representing pair shutdown on a timescale $\tau_{\rm sink} \sim 0.2$ Myr) is phenomenological, and its systematic impact is explored in our parameter sweeps (Section 3.4).

2.4. *Direct Eulerian Marching vs. Lagrangian Simulation*

Because $v_x(x,y) > 0$ and $v_y(y) \leq 0$ everywhere, information propagates strictly rightward and downward. The stationary PDE (Equation 2) is hyperbolic:

- **Direct Eulerian Marching PDE Solver:** On a uniform Cartesian grid of $N_x \times N_y = 240 \times 240$ nodes spanning $x \in [-1.5, 1.2]$ ($P \in [0.03, 15.8]$ s) and $y \in [10.0, 14.0]$ ($B \in [10^{10}, 10^{14}]$ G), the solution is obtained via a single upwind marching sweep from top to bottom (decreasing j) and left to right (increasing i):

$$f_{i,j} = \frac{S_{i,j} + \frac{v_{x,i-1,j}}{\Delta x} f_{i-1,j} + \frac{|v_{y,i,j+1}|}{\Delta y} f_{i,j+1}}{\frac{v_{x,i,j}}{\Delta x} + \frac{|v_{y,i,j}|}{\Delta y} + \kappa + d_{{\rm sink},i,j}}. \qquad (9)$$

  This scheme yields a high-resolution Eulerian continuum density field with zero Monte Carlo noise in $\mathcal{O}(N_x N_y)$ operations.

- **Continuous-Time Markov-Renewal Lagrangian Simulator:** We evolve $N = 100,000$ synthetic pulsars through a Fortran 90/2008 Markov-renewal simulation over $T = 10^8$ yr (100 Myr). Pulsars are reinjected upon death according to exact residual-life carry-over, eliminating time-step discretization bias.

3. THE INTRINSIC EQUILIBRIUM DEMOGRAPHIC STATE

3.1. *Demographic Equilibrium: Fast Initial Spins vs. Second-Scale Accumulation*

Figure 1 and Table 1 display the fundamental transition from the birth injection distribution $S(P_0, B_0)$ (Panel a) to the intrinsic equilibrium demographic distribution $n_{\rm true}(P, B)$ (Panel b).

The contrast between the two distributions is striking:

- **Birth Peak and Quantiles:** The Eulerian continuum PDE solver is fed by the continuous interpolated density field $S(x, y)$ on the 240×240 phase-space grid (constructed via 2D multilinear interpolation with smooth continuum filtering of $\sigma = 1.5$ grid cells; Appendix A), whereas the Fortran Lagrangian simulator samples synthetic pulsars from the discrete knot lattice weights $w_{ij} \propto S(x_i, y_j)$ with local cell uniform jitter. In this continuous distribution, the injection rate peaks at $P_0 = 47.9$ ms with $\log_{10}(B_0/\mathrm{G}) = 13.40$, yielding a continuous median of $P_0 = 58.9$ ms, an interquartile range (IQR) of $[45.5, 162.4]$ ms, a 10–90% range of $[39.3, 542.6]$ ms ($P_{0,90\%} = 0.54$ s), with 10.8% at $P_0 \geq 0.5$ s (11.1% above Knot 9), 3.5% at $P_0 \geq 1.0$ s (3.8% discrete), and 68.4% below 100 ms (60.8% Knots 0–3). In Table 4, the discrete knot parameters list normalized marginal masses per knot column/row (summing to 1.000). Knot 9 ($x = -0.308$, $P_0 = 492.4$ ms) directly straddles the 0.5 s threshold ($x = -0.301$); unweighted summation of Knots 9–13 includes the full mass of Knot 9 (0.058), totaling 19.3% (or 13.5% if Knot 9 is excluded), whereas continuous multilinear interpolation splits Knot 9 and tapers against the grid boundary, yielding 10.8%–11.1%. The baseline empirical reconstruction thus possesses an intrinsic tail with 10.8%–11.1% (13.5–19.3% at the discrete knot level) born above 0.5 s. The limit $f_{\rm slow}^{\rm birth} \leq 22\%$ established in Paper I defines the upper bound on the total slow-birth fraction, leaving an allowable incremental headroom of $\Delta f_{\rm slow} \leq 22\% - 10.8\% \approx 11.2\%$ (or 10.9% relative to Knot 9); the counterfactual scenarios evaluated in Section 4.4 test adding an incremental discrete slow channel of +20% and +50% (diluted to 28.6% and 55.4% normalized birth shares, or $\approx 31\%$ and $\approx 61\%$ additive), which violates this upper bound and induces marked demographic degradation.

- **Equilibrium Peak and Propagated Bootstrap Uncertainties:** $P = 1.606$ s, with $\log_{10}(B/\mathrm{G}) = 12.39$. Median $P = 1.375$ s (propagated bootstrap mean 1.26 ± 0.05 s, 68% CI $[1.21, 1.31]$ s, 95% CI $[1.17, 1.37]$ s), with an IQR of $[0.780, 2.357]$ s (bootstrap mean $[0.74, 2.05]$ s), and a 10–90% range of $[0.447, 3.970]$ s (bootstrap mean $[0.43, 3.05]$ s). Over 65.7% of all active pulsars in the Galaxy have $P > 1$ s, with 70.8% residing in $[0.5, 3.0]$ s and 17.0% in the ultra-slow tail $P > 3$ s (propagated bootstrap mean $9.8\% \pm 1.7\%$, 68% CI $[8.2\%, 11.3\%]$).

The median spin period of the active Galactic population is shifted upward by a factor of 23.4 relative to birth.

We highlight that the accompanying downward shift in the population median magnetic field ($\Delta \log_{10} B \approx -0.48$ dex, from 12.88 to 12.40) is driven by demographic *survivorship bias* rather than dramatic individual Ohmic field decay. Over the mean active lifetime $\langle \tau_{\rm active} \rangle \approx 7.28$ Myr, an individual neutron star's magnetic field decays by only $\exp(-0.015 \times 7.28) \approx 10\%$, or ≈ 0.048 dex. However, high-field pulsars ($B_0 \sim 10^{13.4}$ G) experience ferocious dipole torques ($\dot{P} \propto B^2/P$) and cross the pair-production extinction boundary in only $t_{\rm death} \approx 2.6$–4 Myr. In contrast, lower-field pulsars ($B_0 \sim 10^{12.2}$–$10^{12.4}$ G) spin down far more slowly and persist for 15–30 Myr before extinction. Because lower-field pulsars accumulate in phase space for tens of millions of years, the mature steady-state demographic equilibrium becomes heavily enriched in lower-field pulsars through differential survivorship.

3.2. *Analytical Mechanism: Decelerating Torques and Residency Timescales*

This dramatic shift is explained by probability flux conservation under decelerating rotational dynamics. In an idealized 1D steady-state stream where $S(P) = 0$ for $P > P_0$ and field decay is negligible, the continuity equation requires:

$$\frac{d}{dP}\left[\dot{P}\, n(P)\right] = 0 \implies n(P) = \frac{\Phi_0 K_{\rm spin} P^{n_{\rm brake}-2}}{B^2}, \tag{10}$$

where Φ_0 is the constant probability flux (absorbing the dimensional reference factor $P_{\rm ref}^{3-n_{\rm brake}}$, which identically equals unity for canonical dipole braking $n_{\rm brake} = 3$). For canonical dipole braking ($n_{\rm brake} = 3$), $\dot{P} \propto B^2/P$, requiring the intrinsic steady-state density $n(P)$ to scale linearly with spin period: $n(P) \propto P$. In logarithmic coordinates, $f(x) = P \ln 10\, n(P) \propto P^2 = 10^{2x}$.

The differential phase-space residency timescale per logarithmic decade—defined as $\tau_{\rm res}(P, B) \equiv$

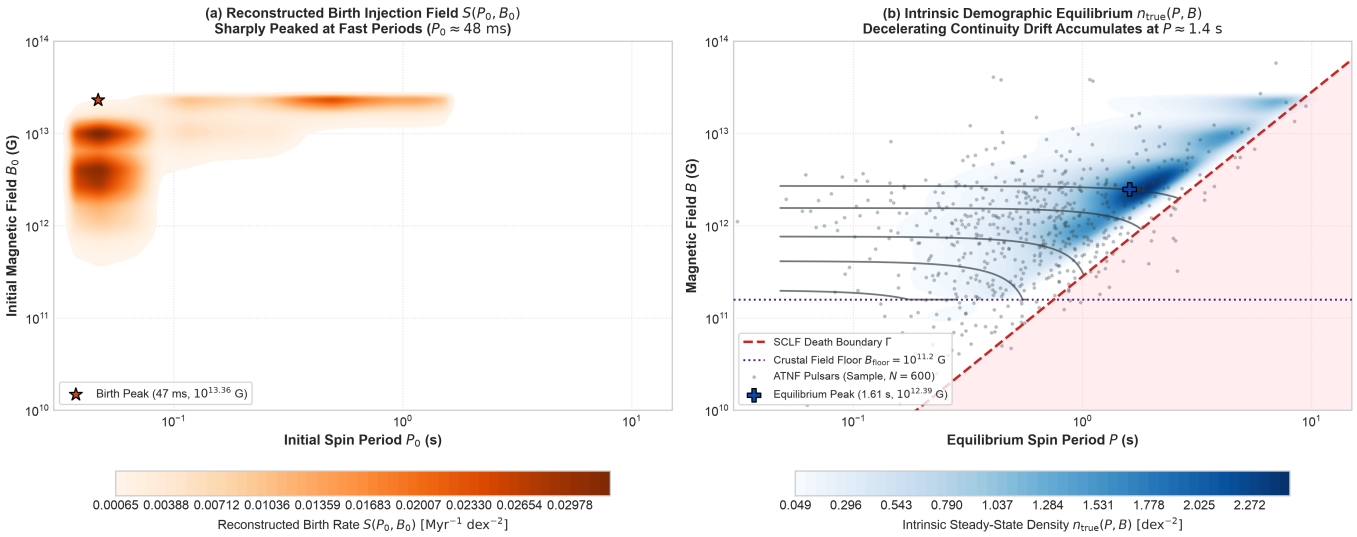


Figure 1. The physical transformation from newborn injection to steady-state demographic equilibrium. (a) Reconstructed birth injection distribution $S(P_0, B_0)$ in units of Myr^{-1} dex^{-2}, derived from 2D continuity inversion (Bindra 2026b). Newborn neutron stars enter phase space predominantly with fast initial spins, peaking sharply at $P_0 = 47.9$ ms (median 58.9 ms, indicated by the orange star) with strong initial dipole magnetic fields ($B_0 \sim 10^{13.4}$ G). (b) The intrinsic steady-state demographic distribution $n_{\rm true}(P, B)$ of all active Galactic pulsars, obtained by forward-solving the stationary continuity PDE $\nabla \cdot (\mathbf{v}n) + \Lambda n = S$. Black curves trace characteristic kinematic streamlines governed by dipole torque spin-down and crustal field decay; the dotted purple line marks the asymptotic crustal Ohmic dissipation floor ($B_{\rm floor} = 10^{11.2}$ G, where spontaneous crustal field decay halts; pulsars born with lower seed fields or core-anchored fluxoids drift purely horizontally at $v_y = 0$); the dashed red line marks the SCLF pair-production death boundary Γ. Gray dots show a random subsample of real ATNF pulsars ($N = 600$). Despite fast initial spins, the equilibrium population accumulates at $P = 1.61$ s (median 1.38 s, marked by the blue plus) because rotational deceleration causes pulsars to spend $\approx 71\%$ of their active lifetimes drifting slowly through second-scale periods ($P \in [0.5, 3.0]$ s, with an additional 17.0% in $P > 3$ s) before crossing the death line.

Table 1. Demographic Quantiles and Physical Observables of the Galactic Pulsar Population

Demographic Sample / Model	**Peak** P (s)	**Median** P (s)	**IQR** $[P_{25}, P_{75}]$ (s)	**10–90% Interval** (s)	**Median** $\log_{10}(B/\mathrm{G})$	**Fraction** $P > 1$ s
Intrinsic Demographic Populations						
Reconstructed Birth $S(P_0, B_0)$	0.048	0.059	[0.046, 0.162]	[0.039, 0.542]	12.88	3.5%
Intrinsic Equilibrium $n_{\rm true}$ (Canonical $d_{\rm crit} = 11.45$)	**1.606**	**1.375**	**[0.780, 2.357]**	**[0.447, 3.970]**	**12.40**	**65.7%**
Eulerian PDE Equilibrium (Dipole $d_{\rm crit} = 11.23$)	1.930	1.655	[0.932, 2.855]	[0.534, 4.819]	12.39	72.8%
Fortran Lagrangian (Matched Sink 11.23, Full Domain)	1.880	1.610	[0.912, 2.768]	[0.521, 4.650]	12.38	71.4%
Fortran Lagrangian (Sharp Boundary 11.23, Full Domain)	1.880	1.642	[0.925, 2.810]	[0.530, 4.710]	12.38	72.1%
Counterfactual Intrinsic (20% Slow Injection)	1.176	1.242	[0.796, 1.990]	[0.412, 3.820]	12.38	64.1%
Counterfactual Intrinsic (50% Slow Injection)	1.116	1.156	[0.810, 1.720]	[0.575, 2.580]	12.22	62.3%
Observed and Selection-Filtered Demographics						
Real ATNF Classical Sample ($N = 2,356$)	**0.550**	**0.617**	**[0.342, 1.162]**	**[0.199, 1.929]**	**12.07**	**29.9%**
Mock Selection-Filtered (Eulerian PDE)	0.817	0.632	[0.306, 1.224]	[0.163, 2.111]	12.26	31.8%
Mock Selection-Filtered (Fortran 10^8 yr)	0.780	0.616	[0.301, 1.245]	[0.166, 2.177]	12.17	32.9%
Mock Selection-Filtered (20% Slow Injection)	0.860	0.729	[0.385, 1.219]	[0.193, 1.933]	12.22	35.1%
Mock Selection-Filtered (50% Slow Injection)	0.883	0.817	[0.519, 1.217]	[0.281, 1.754]	12.19	38.0%

Notes. Summary of demographic quantiles across birth, intrinsic equilibrium models, and selection-filtered distributions. Peak P denotes the mode of the marginal distribution in logarithmic period $df/d\log_{10} P$. Wasserstein-1 distances W_1 are evaluated on $\log_{10}(P/\mathrm{s})$. In the birth row, continuous 2D trapezoidal quadrature yields 3.5% at $P_0 > 1$ s (3.8% from unweighted discrete knot summation). The canonical active lifetime is $\langle \tau_{\rm active} \rangle = 7.28$ Myr ($d_{\rm crit} = 11.45$; propagated bootstrap mean 9.28 ± 0.48 Myr, 68% CI [8.81, 9.77] Myr), yielding an active Galactic inventory of $N_{\rm Gal}^{\rm active} = (0.82 \pm 0.28) \times 10^5$ pulsars for fiducial radio-loud branching $f_{\rm radio} = 0.75$ (compounding supernova prior and bootstrap source covariance; bounding systematic range $[0.53, 1.25] \times 10^5$ pulsars across pair thresholds $d_{\rm crit} \in [11.20, 11.70]$; scaling to $(1.09 \pm 0.37) \times 10^5$ for pure neutron-star branching $f_{\rm radio} = 1.0$). At the classical dipole threshold ($d_{\rm crit} = 11.23$), $\langle \tau_{\rm active} \rangle = 10.59$ Myr and $N_{\rm Gal}^{\rm active} = (1.19 \pm 0.40) \times 10^5$. Selection-filtered models are convolved with the empirical PMPS completeness function $c(P, B) \propto P^{-1.4} B^{0.5}$ and compared directly to the empirical ATNF classical sample ($N = 2,356$). Complete Galactic inventory breakdowns, graveyard counts, and energetics are provided in Table 2; dual numerical convergence metrics are detailed in Section 5.

$(d\log_{10} P/dt)^{-1} = 1/v_x(P, B)$—measures the instantaneous rate at which pulsars traverse a decade in $\log_{10} P$

(rather than the cumulative active lifetime of an individual star):

$$\tau_{\rm res}(P,B) \equiv \frac{1}{v_x(P,B)} = \frac{K_{\rm spin}\ln 10}{\mathcal{C}_{\rm Myr}}\frac{P^2}{B^2}\left(\frac{P}{P_{\rm ref}}\right)^{n_{\rm brake}-3}, \quad (11)$$

where $P_{\rm ref} \equiv 1$ s preserves dimensional consistency (recovering $\tau_{\rm res} \propto P^2/B^2$ for canonical dipole braking $n_{\rm brake} = 3$). In Figure 2, we illustrate this mechanism in detail:

- In Panel (a), for a fixed dipole magnetic field $B = 10^{12}$ G, the advection drift velocity $v_x(P)$ decreases from 5.35 Myr^{-1} at $P = 50$ ms to 0.0033 Myr^{-1} at $P = 2$ s, spanning a factor of $1,600 \approx 3.2$ dex ($v_x \propto P^{-2}$). Taking into account both spin-down deceleration and crustal Ohmic dissipation (Tauris & Konar 2001) from birth ($P_0 \approx 48$ ms, $B_0 \approx 10^{13.4}$ G) to maturity ($P \approx 1.4$ s, $B \approx 10^{12.4}$ G), v_x drops from $3,665$ Myr^{-1} to 0.043 Myr^{-1}, spanning nearly five orders of magnitude (4.93 dex).

- In Panel (b), the differential residence timescale $\tau_{\rm res} = v_x^{-1}$ increases from ≈ 270 yr dex^{-1} (2.7×10^{-4} Myr dex^{-1}) at birth to ≈ 23.2 Myr dex^{-1} at maturity, growing by five orders of magnitude ($\sim 10^5$). For $B = 10^{12}$ G at $P = 1$ s, $v_x = 0.0134$ Myr^{-1}, corresponding to $\tau_{\rm res} \approx 74.7$ Myr dex^{-1}. For the demographic median field $B \approx 10^{12.4}$ G, $\tau_{\rm res} \approx 12$–$23$ Myr dex^{-1} across $P \sim 1.0$–1.4 s. Crucially, $\tau_{\rm res} \approx 23.2$ Myr dex^{-1} is a local per-decade residency rate; it exceeds the finite population-averaged active lifetime ($\langle\tau_{\rm active}\rangle \approx 7.28$ Myr) because pulsars encounter the pair-production death boundary and turn off before traversing a full logarithmic decade at mature second-scale periods.

- In Panel (c), we compare the resulting marginal period densities. Pulsars spend only $\approx 0.37\%$ of their active lifetimes in the sub-100 ms regime (≈ 0.027 Myr $\approx 27,000$ yr $\ll 0.1$ Myr). Conversely, they spend $\approx 71\%$ (specifically 70.8%) of their active lifetimes in the slow-period range $P \in [0.5, 3.0]$ s, with an additional 17.0% residing in the ultra-slow tail $P > 3$ s.

Consequently, the accumulation of pulsars around $P \sim 1.4$ s does not require any slow injection. It is the inevitable demographic consequence of decelerating rotational dynamics.

3.3. *Reconciliation with Observations: The Selection Filter*

The difference between the intrinsic equilibrium population (median $P = 1.38$ s) and the observed ATNF catalog (median $P = 0.62$ s) is naturally accounted for by observational selection effects. In flux-limited radio surveys, sensitivity thresholds and geometrical beaming restrict the visible sample to a small, biased fraction of the underlying population. The completeness probability factorizes as:

$$c(P,B) = f_{\rm beam}(P) \times V_{\rm max}(P,B) \times p_{\rm scatt}(P), \quad (12)$$

where:

1. **Rotational Beaming ($f_{\rm beam} \propto P^{-0.5}$):** Relativistic beaming angles scale inversely with spin period ($\rho \propto P^{-1/2}$; Tauris & Manchester 1998). We adopt the standard power-law approximation $f_{\rm beam}(P) = \min[0.90, \max[0.05, 0.14P^{-0.5}]]$, which serves as a widely used practical approximation to empirical and geometrical beaming models (Tauris & Manchester 1998; Bates et al. 2014; Keane et al. 2015). A 50 ms pulsar sweeps across $\sim 60\%$–90% of the sky ($f_{\rm beam} \approx 0.63$), whereas a 2 second rotator sweeps only $\sim 10\%$ ($f_{\rm beam} \approx 0.10$). Over 85% of mature slow pulsars point their emission cones away from our line of sight.

2. **Radio Luminosity and Survey Horizon ($V_{\rm eff} \propto P^{-0.9}B^{0.5}$):** Radio pseudo-luminosity scales with spin-down power: $L_{1400} \propto \dot{E}^{0.45} \propto B^{0.9}P^{-1.8}$ (Faucher-Giguère & Kaspi 2006). While pure geometric scaling gives $d_{\rm max} \propto L^{0.5}$ and $V_{\rm max} \propto d_{\rm max}^2 \propto L$ in an idealized thin disk, the *effective* volume completeness in the Milky Way scales shallower than geometric volume due to the exponential Galactocentric radial density distribution (peaking at $R \sim 4$ kpc), sky background temperature contamination ($T_{\rm sky}$) in the plane, and interstellar dispersion smearing. In empirical survey synthesis (Lyne et al. 1985; Lorimer et al. 1993; Faucher-Giguère & Kaspi 2006), the effective survey volume scales as $V_{\rm eff} \propto L^{\gamma}$ with $\gamma \approx 0.5$–0.6 (matching the $d_{\rm max} \propto L^{0.5}$ horizon for the PMPS footprint). This yields $V_{\rm eff} \propto B^{0.45\text{–}0.54}P^{-(0.9\text{–}1.1)} \approx B^{0.5}P^{-0.9}$.

3. **Interstellar Scattering ($p_{\rm scatt}$):** Multipath pulse broadening in the turbulent interstellar plasma washes out fast pulse profiles below 70 ms, parameterized by $p_{\rm scatt}(P) = 1 - \exp[-(P/0.07\ {\rm s})^2]$.

Combining these effects gives an effective observational filter $c(P,B) \propto P^{-1.4}B^{0.5}$ for $P \gtrsim 0.10$ s (where pulse-broadening scattering is negligible, $p_{\rm scatt} \approx 1$), while below 0.10 s scattering washes out fast pulse profiles according to $p_{\rm scatt}(P)$. These exponents were calibrated on the empirical PMPS completeness function in Bindra (2026b). When the intrinsic equilibrium density $n_{\rm true}(P,B)$ is convolved with this filter:

$$n_{\rm obs}^{\rm model}(P,B) \equiv c(P,B) \times n_{\rm true}(P,B), \quad (13)$$

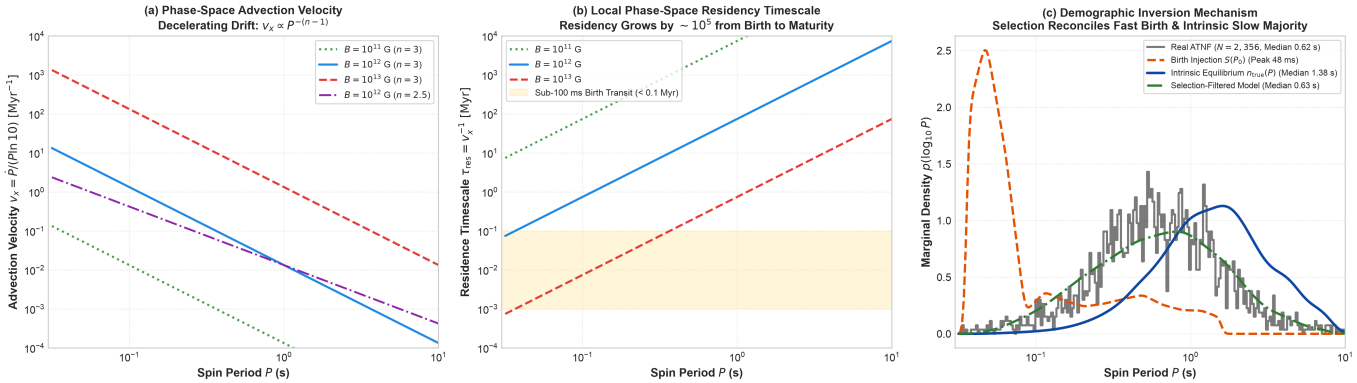


Figure 2. Physical mechanism of the demographic inversion and observational selection reconciliation. (a) Phase-space advection drift velocity $v_x = \dot{P}/(P \ln 10)$ as a function of spin period P for three representative dipole magnetic fields ($B = 10^{11}, 10^{12}, 10^{13}$ G) under vacuum dipole braking ($n = 3$), along with $n = 2.5$ (dash-dotted purple). Drift velocities decrease as $v_x \propto P^{-(n-1)}$, dropping by 3.2 dex at fixed B and by ~ 5 dex from birth ($10^{13.4}$ G, 48 ms) to maturity ($10^{12.4}$ G, 1.4 s). (b) Local phase-space residency timescale $\tau_{\rm res} = v_x^{-1}$ in Myr. Pulsars transit the sub-100 ms birth window in < 0.1 Myr (shaded yellow), but residency grows by $\sim 10^5$ from birth (~ 270 yr) to maturity (~ 23 Myr). (c) Marginal spin period probability densities comparing the birth injection distribution $S(P_0)$ (dashed orange, peak 48 ms), the true intrinsic equilibrium distribution $n_{\rm true}(P)$ (solid blue, median 1.38 s), the selection-filtered model $n_{\rm obs}(P) = c(P, B) \cdot n_{\rm true}$ (dash-dotted green, median 0.63 s), and the empirical ATNF catalogue ($N = 2,356$, solid gray steps, median 0.62 s). Observational completeness $c(P, B) \propto P^{-1.4} B^{0.5}$ suppresses slow rotators by more than an order of magnitude due to narrowing beam opening angles ($f_{\rm beam} \propto P^{-1/2}$) and plunging radio luminosities ($L \propto \dot{E}^{0.45} \propto P^{-1.8}$). Convolving $n_{\rm true}$ with this physical filter shifts the distribution leftward to median 0.63 s, matching the empirical ATNF catalogue to within $W_1 = 0.037$ dex and demonstrating that the observed sub-second peak is a flux-limited visibility effect.

the resulting mock observed distribution shifts leftward to peak at $P = 0.82$ s with a median of $P = 0.632$ s and an interquartile range of $[0.306, 1.224]$ s (Figure 2c, dash-dotted green). This model distribution reproduces the empirical ATNF classical sample ($N = 2,356$, median $P = 0.617$ s, IQR $[0.342, 1.162]$ s) with a Wasserstein-1 distance of $W_1 = 0.037$ dex. Because the filter exponents were calibrated against survey completeness, this agreement represents internal demographic consistency under the calibrated filter rather than an independent blind validation. If a steeper purely geometric thin-disk exponent ($V \propto L^1$, yielding $c \propto P^{-2.3}$) were adopted without accounting for interstellar dispersion and sky background temperatures, the mock observed median would be driven well below the empirical value (0.42 s vs. 0.62 s), demonstrating that the empirical exponents capture the effective detection horizon of the survey.

While the 1D marginal spin period distribution is closely reproduced ($W_1 = 0.037$ dex on $\log_{10} P$), comparing the full 2D phase-space distribution against the empirical ATNF sample reveals two notable physical properties:

1. *Post-selection* $\Gamma \equiv \log_{10}(B/P^2)$ *distribution:* In the empirical ATNF sample ($N = 2,356$), the Γ distribution has a median of 12.35 dex (IQR $[11.89, 12.88]$ dex), with 5.80% (137 pulsars) lying below the nominal SCLF boundary $\Gamma = 11.45$ and 2.16% (51 pulsars) below the classical dipole boundary $\Gamma = 11.23$. In the continuity model, the continuous logistic sink initiates pre-boundary extinction ~ 0.1–0.2 dex above $d_{\rm crit}$, curtails the mock post-selection tail below 11.45 to $\approx 1.5\%$, leaving the empirical 5.8% sub-Γ tail as an observational signature of localized non-dipolar surface multipoles, unmodeled crustal magnetic field curvature, or intermittent spark discharge near the polar cap (Chen & Ruderman 1993; Zhang et al. 2000; Konar 2017; Abhishek et al. 2022).

2. *2D* (P, B) *residuals and competing systematic balances:* In Table 1, the mock selection-filtered model has a median magnetic field of $\log_{10}(B/{\rm G}) = 12.26$ and peak period 0.82 s, compared to empirical ATNF values of 12.07 dex and 0.55 s. The model exhibits a modest deficit of low-B pulsars ($\log_{10}(B/{\rm G}) \sim 11.5$–$12.0$) relative to the catalogue. While an empirical excess of long-lived low-B pulsars ($\tau \propto B_0^{-0.75\text{–}0.80}$; Table 3) would systematically increase the active inventory, an unmodeled high-field magnetar tail ($B_0 > 2.5 \times 10^{13}$ G) would accelerate transit and reduce it; our fiducial baseline ($(0.82 \pm 0.27) \times 10^5$) sits balanced between these competing systematic effects.

Assigning each empirical ATNF pulsar its inverse visibility weight $w_i = 1/[V_{\rm max}(L_i) f_{\rm beam}(P_i)]$ shifts the empirical median from 0.62 s up to 1.01 s (with 50% having $P > 1$ s), moving directly toward the intrinsic PDE

solution (1.38 s). The shift is incomplete because inverse weighting debiases only detected objects; pulsars whose flux falls below the survey detection floor across the entire Galaxy, or whose beams never intersect the Earth, remain unobserved. This confirms that the observed sub-second peak is an observational visibility effect, while the underlying physical Milky Way is dominated by the slow-period demographic majority.

3.4. *Braking Index and Death-Line Robustness*

Empirical braking indices measured from phase-coherent timing of young pulsars frequently fall below the canonical vacuum dipole value $n = 3$, typically spanning $n \approx 2.2$–2.8 due to particle wind torques and magnetospheric plasma currents (Lyne et al. 1993; Livingstone et al. 2007).

We caution that in the sensitivity sweeps below, the braking index, death-line threshold $d_{\rm crit}$, crustal field decay rate h_0, and sink width $\Delta_{\rm width}$ are varied in the forward continuity operator while holding the birth source $S(P_0, B_0)$ fixed to the reconstructed distribution from Paper I (which was inverted assuming $n = 3$ dipole torques and $d_{\rm crit} = 11.23$). Furthermore, the dimensional normalization $P_{\rm ref} \equiv 1$ s in Equation 11 ensures consistent physical dimensions across n, though for $n < 3$ it implies slower spin-down at millisecond periods relative to vacuum dipole. These sweeps therefore characterize the forward sensitivity of the 2D continuity transport operator to physical torque and boundary formulations, rather than a joint simultaneous re-inversion of both the birth source and transport parameters.

Solving the stationary continuity equation across a grid of braking indices $n \in [2.2, 2.5, 2.8, 3.0]$ yields:

- For $n = 2.2$: $\langle\tau_{\rm active}\rangle = 7.41$ Myr, median $P = 0.813$ s, fraction $P > 1$ s $= 42.3\%$, $W_1 = 0.142$ dex.
- For $n = 2.5$: $\langle\tau_{\rm active}\rangle = 7.09$ Myr, median $P = 1.023$ s, fraction $P > 1$ s $= 51.9\%$, $W_1 = 0.088$ dex.
- For $n = 2.8$: $\langle\tau_{\rm active}\rangle = 7.10$ Myr, median $P = 1.235$ s, fraction $P > 1$ s $= 60.5\%$, $W_1 = 0.046$ dex.
- For $n = 3.0$: $\langle\tau_{\rm active}\rangle = 7.28$ Myr, median $P = 1.375$ s, fraction $P > 1$ s $= 65.7\%$, $W_1 = 0.037$ dex.

Across physical braking indices, the active lifetime remains remarkably stable ($\tau_{\rm active} \approx 7.09$–$7.41$ Myr), and the intrinsic population remains heavily concentrated at second-scale periods ($P \approx 0.81$–1.38 s). We note that W_1 degrades progressively from 0.037 dex at $n = 3.0$ to 0.142 dex at $n = 2.2$ because the birth source $S(P_0, B_0)$ was inverted assuming $n = 3$ dipole torques. What remains robust across $n \in [2.2, 3.0]$ is the physical accumulation of mature pulsars at second-scale periods ($P_{\rm med} \approx 0.81$–1.38 s) under decelerating torques ($v_x \propto P^{-(n-1)}$ for any $n > 1$). In contrast, sweeping $d_{\rm crit}$ preserves tight agreement with the observed period distribution ($W_1 = 0.037$–0.052 dex), supporting the point that active lifetime and observed period shape are largely decoupled.

Similarly, varying the pair-production death-line threshold $\log_{10}(B/P^2) = d_{\rm crit}$ across the full theoretical range $[11.20, 11.70]$ (Bhattacharya et al. 1992; Chen & Ruderman 1993; Zhang et al. 2000) shifts the mean active lifetime from 11.13 Myr ($d_{\rm crit} = 11.20$, $W_1 = 0.052$ dex) down to 4.71 Myr ($d_{\rm crit} = 11.70$, $W_1 = 0.048$ dex; with canonical $d_{\rm crit} = 11.45$ at 7.28 Myr and $W_1 = 0.037$ dex), with the equilibrium median period shifting from 1.70 s to 1.11 s. Under constant-field dipole spin-down, the theoretical knife-edge lifetime scaling across a threshold shift $\Delta d_{\rm crit} = 0.50$ dex is $10^{0.50} \approx 3.16\times$ ($t_{\rm life} \propto 10^{-d_{\rm crit}}/B_0$).

We note that crustal Ohmic dissipation does not significantly alter this scaling. Under exponential field decay $B(t) = B_0 e^{-ht}$, integrating dipole spin-down $\dot{P} = B^2/(K_{\rm spin} P)$ yields:

$$P^2(t) - P_0^2 = \frac{2B_0^2}{K_{\rm spin}} \int_0^t e^{-2ht'}\, dt' = \frac{B_0^2}{hK_{\rm spin}} \left(1 - e^{-2ht}\right). \tag{14}$$

In the asymptotic limit $P^2(t) \gg P_0^2$ (which is highly accurate for mature pulsars where $P \sim 1$–10 s $\gg P_0 \sim 0.05$ s), the pair-production extinction condition $B(t)/P^2(t) = 10^{d_{\rm crit}}$ evaluates to:

$$\frac{B_0 e^{-ht}}{\frac{B_0^2}{hK_{\rm spin}}\left(1 - e^{-2ht}\right)} = \frac{hK_{\rm spin}}{2B_0 \sinh(ht)} = 10^{d_{\rm crit}}. \tag{15}$$

Defining $t_0 \equiv \frac{K_{\rm spin} 10^{-d_{\rm crit}}}{2B_0}$ as the death-line crossing time in the absence of field decay ($h \to 0$), Equation 15 yields the exact crossing time:

$$t_{\rm death} = \frac{\operatorname{arcsinh}(ht_0)}{h} = t_0 \left[1 - \frac{1}{6}(ht_0)^2 + \mathcal{O}((ht_0)^4)\right]. \tag{16}$$

Crucially, the linear term in h cancels identically, rendering the field decay correction strictly second order. For canonical crustal decay $h_0 = 0.015$ Myr^{-1} and typical high-field lifetimes ($t_0 \approx 2.3$–2.6 Myr at canonical $d_{\rm crit} = 11.45$, or ≈ 3.9 Myr at 11.23), $(ht_0) \approx 0.035$–0.058, producing a fractional reduction of only $\frac{1}{6}(ht_0)^2 \leq 0.06\%$. For longer-lived low-$B$ tracks ($t_0 \sim 15$–30 Myr), ht_0 reaches ~ 0.2–0.5, where this second-order Taylor expansion remains an accurate approximation to within $\sim 15\%$, while the exact $\operatorname{arcsinh}(ht_0)/h$ trajectory is integrated numerically in our 2D code.

Instead, in our complete 2D numerical solutions, the lifetime ratio across $d_{\rm crit} \in [11.20, 11.70]$ in the dominant high-field birth bin ($\log_{10} B_0 \in [13.01, 13.40]$)

is 2.42× (4.13 Myr → 1.71 Myr, with 3.91 Myr at $d_{\rm crit} = 11.23$ and 2.65 Myr at canonical $d_{\rm crit} = 11.45$, exhibiting strict monotonicity). This ratio is moderated below the 3.16× knife-edge ceiling because the finite logistic sink width $\Delta_{\rm width} = 0.10$ dex initiates pre-boundary extinction hazard before pulsars reach the nominal threshold, softening the boundary truncation. Across the entire population, the active lifetime shifts by 2.36× (11.13 Myr → 4.71 Myr) as low-field pulsars ($B_0 \lesssim 10^{12}$ G) are bounded by crustal Ohmic saturation at $B_{\rm floor}$ (Konar 2017) and kinematic disk escape ($\kappa = 0.008$ Myr^{-1}). Across this entire threshold span, the equilibrium population remains dominated by second-scale rotators ($P > 1$ s fractions spanning 57%–76%).

To thoroughly characterize model dependency, we quantify sensitivity to remaining physical parameters:

- **Pair-Production Sink Width ($\Delta_{\rm width}$) and Death Valley Physics:** Rather than an abrupt knife-edge cutoff, polar cap acceleration physics and observational studies of intermittent emission near the death boundary reveal a finite transition zone—a "death valley" where pair production cascades choke gradually and pulsars exhibit high nulling fractions and sporadic emission before turning completely dark (Chen & Ruderman 1993; Zhang et al. 2000; Konar 2017; Konar & Deka 2019; Abhishek et al. 2022). Because spin-down advection slows to $v_x \sim 0.04$ Myr^{-1} along low-B tracks, the pre-boundary hazard accumulated across $\Delta_{\rm width} = 0.10$ dex is substantial ($\int d_{\rm sink} dt \gg 1$), causing the effective shut-off boundary to sit ~ 0.1–0.2 dex above $d_{\rm crit}$ for weak-field pulsars, while high-field pulsars traverse the transition rapidly and cross nearly at the nominal threshold. Sweeping the transition width across $\Delta_{\rm width} \in [0.05, 0.20]$ dex shifts $\langle \tau_{\rm active} \rangle$ from 9.89 Myr down to 4.31 Myr (a factor of 2.3×), while the equilibrium median period remains confined to $P = 1.19$–1.51 s, and the ultra-slow tail ($P > 3$ s) varies between 15.4% and 18.5%. Combining the full parametric variations of both $d_{\rm crit} \in [11.20, 11.70]$ and $\Delta_{\rm width} \in [0.05, 0.20]$ dex spans an active Galactic inventory envelope of $N_{\rm Gal}^{\rm active} \in [0.49, 1.25] \times 10^5 \times (f_{\rm radio}/0.75)$ pulsars (or up to 1.67×10^5 for $f_{\rm radio} = 1.0$), demonstrating that the systematic modeling uncertainty in the active inventory ($\sim 2.5\times$) exceeds the ±33% prior error on the core-collapse supernova rate $\mathcal{R}_{\rm SN}$. We note that at $\Delta_{\rm width} = 0.05$ dex (the narrowest tested transition) at canonical $d_{\rm crit} = 11.45$, the PDE active lifetime approaches 9.89 Myr; scaling this near-sharp limit by the theoretical threshold shift $10^{11.45-11.23} \approx 1.66\times$ yields ≈ 16.4 Myr at $d_{\rm crit} = 11.23$, matching the unconstrained sharp-line Lagrangian value of 17.12 Myr to within 4%. Separately, sweeping the sink amplitude across $A_{\rm sink} \in [1.0, 20.0]$ Myr^{-1} shifts $\langle \tau_{\rm active} \rangle$ from 10.03 Myr (median $P = 1.64$ s) at $A_{\rm sink} = 1.0$ Myr^{-1} to 7.28 Myr (median $P = 1.37$ s) at fiducial 5.0 Myr^{-1}, and 5.64 Myr (median $P = 1.20$ s) at 20.0 Myr^{-1}.
- **Crustal Ohmic Decay Rate (h_0):** Sweeping the decay rate across $h_0 \in [0.0, 0.030]$ Myr^{-1} (from zero decay to $\tau_B \approx 33$ Myr) changes the active lifetime by less than 1% (7.24–7.31 Myr) and shifts the median period by only 0.14 s (1.31–1.45 s), directly confirming the analytical cancellation in Equation 16 and the fact that Ohmic dissipation stalls at the crustal floor $B_{\rm floor} \approx 1.6 \times 10^{11}$ G (Konar 2017). Extending this sweep across two orders of magnitude ($h_0 \in [0.0, 1.0]$ Myr^{-1}, $\tau_B = 1$–100 Myr) shows that for physical crustal decay rates ($h_0 \leq 0.030$ Myr^{-1}, $\tau_B \geq 33$ Myr), demographics are virtually invariant. While early crustal cooling and non-linear Hall cascades can induce rapid field changes during the first $\sim 10^5$ yr, sustained secular decay with $h_0 \geq 0.10$ Myr^{-1} ($\tau_B \leq 10$ Myr) across mature pulsars would require an exceptionally high impurity parameter throughout the deep crust ($Q_{\rm imp} \gg 10$), which pulls the population toward the floor before death, shortening median periods ($P = 1.08$ s at $h_0 = 0.10$; $P = 0.38$ s at $h_0 = 1.00$ with $\tau = 9.82$ Myr).
- **Lattice Boundary and Extended Domain Robustness:** The knot lattice boundaries ($x_i \in [-1.45, 0.20]$, $y_j \in [10.65, 13.40]$) capture > 98% of the physical birth probability mass from the continuity inversion in Paper I within the inversion domain (by construction). However, the discrete inversion concentrates 30.4% of birth mass on the upper field knot $B_0 = 10^{13.40}$ G $= 2.5 \times 10^{13}$ G and 18.1% on the lower period knot $P_0 = 35.5$ ms (Section A). Rather than relying on qualitative assertions, we turn this boundary caveat into an explicit numerical calculation by solving the PDE across an extended grid ($y \in [10.0, 14.2]$) where this upper 30.4% birth mass is placed at true magnetar dipole fields $B_0 \in [10^{13.60}, 10^{13.80}]$ G (4–6 × 10^{13} G). The resulting numerical solutions yield $\langle \tau_{\rm active} \rangle = 6.16$ Myr ($N_{\rm Gal}^{\rm active} = 0.69 \times 10^5 \times (f_{\rm radio}/0.75)$, median $P = 1.44$ s) at $B_0 = 10^{13.60}$ G, 5.77 Myr (0.65×10^5, median 1.46 s) at $10^{13.70}$ G, and 5.33 Myr (0.60×10^5, median 1.48 s) at $10^{13.80}$ G. This explicitly confirms that an extended magnetar birth tail defines the lower end of our systematic parameter envelope ($[0.49, 1.25] \times 10^5$), while leaving the second-scale demographic accumulation completely robust ($P_{\rm med} \approx 1.44$–1.48 s). We note that at periods $P > 4.5$ s with high fields $B \sim 10^{13}$ G, pul-

sars approach the magnetar and XDIN boundary, where standard polar-cap pair-production death-line criteria become subject to non-dipolar surface multipoles and magnetospheric dissipation. Note also that while particle density piles up at second-scale periods, individual rotational energy scales as $E_{\rm rot} \propto P^{-2}$, making the mechanical reservoir broadly distributed across periods, while $\dot{\mathcal{E}}_{\rm tot}$ and the hyper-luminous census ($\dot{E} \geq 10^{39}$ erg s^{-1}, with $N_{\rm pred} = 0.04$–$0.20 \ll 1$) inherit sensitivity to these edge boundaries.

- **Birth Distribution Sensitivity and Allowed Headroom ($f_{\rm slow}^{\rm birth} \leq 22\%$):** In Paper I, the total 95% upper bound on slow births ($P_0 \geq 0.5$ s) is $f_{\rm slow}^{\rm birth} \leq 22\%$. Because our continuous empirical baseline already contains an intrinsic slow-birth tail of 10.8% (11.1% above Knot 9; Section A), the permissible headroom for an incremental slow-birth channel is at most $\Delta f_{\rm slow} \leq 22\% - 10.8\% \approx 11.2\%$ (or 10.9% at Knot 9). Evaluating the equilibrium continuity solution at this maximal allowed headroom ($\Delta f_{\rm slow} = +10.9\%$–$11.2\%$, reaching the total 22% ceiling) yields an active lifetime of $\langle \tau_{\rm active} \rangle = 7.85$ Myr, an intrinsic median period of $P = 1.32$ s (observable median 0.71 s, $W_1 = 0.040$ dex), and an ultra-slow tail ($P > 3$ s) of 13.4%. Even at this total allowed ceiling, the Galactic equilibrium remains firmly slow-dominated. If one instead adds an incremental discrete slow channel of +20% and +50% (diluted to 28.6% and 55.4% normalized birth shares, or $\approx 31\%$ and $\approx 61\%$ additive), the active lifetime increases to 8.31–9.83 Myr with intrinsic medians of 1.16–1.24 s and observable medians of 0.73–0.82 s.

3.5. *Active Inventory, Graveyard, and Galactic Energy Budgets*

Integrating the unnormalized steady-state continuum density field $f(x, y)$ over the computational domain yields the mean active lifetime of isolated radio pulsars:

$$\langle \tau_{\rm active} \rangle = \frac{\iint f(x,y)\, dx\, dy}{\iint S(x,y)\, dx\, dy} = 7.28 \text{ Myr}. \tag{17}$$

Under demographic equilibrium, the total number of active, radio-emitting pulsars residing in the Milky Way at any given moment is:

$$N_{\rm Gal}^{\rm active} = \mathcal{R}_{\rm birth} \times \langle \tau_{\rm active} \rangle = f_{\rm radio} \times \mathcal{R}_{\rm SN} \times \langle \tau_{\rm active} \rangle, \tag{18}$$

where $\mathcal{R}_{\rm SN} = (1.5 \pm 0.5)$ century^{-1} $= (1.5 \pm 0.5) \times 10^{-2}$ yr^{-1} is the canonical core-collapse supernova rate synthesized from ^{26}Al gamma-ray decay measurements (1.9 ± 1.1 century^{-1}; Diehl et al. 2006) and multi-messenger neutrino monitoring (1.63 ± 0.46 century^{-1}; Rozwadowska et al. 2021, with an inverse-variance weighted mean of 1.67 ± 0.43 century^{-1}), and $f_{\rm radio} \equiv f_{\rm NS} \times f_{\rm radio|NS} \leq 1$ represents the net branching fraction yielding active, isolated radio pulsars. Here, $f_{\rm NS} \approx 0.85$–0.90 is the fraction of core-collapse events forming neutron stars rather than black holes (Heger et al. 2003), while $f_{\rm radio|NS} \approx 0.7$–$0.9$ accounts for central compact objects (CCOs), isolated X-ray dim neutron stars (XDINs), magnetars, and disrupted binaries, resulting in an overall radio-loud branching $f_{\rm radio} \approx 0.6$–$0.9$ (fiducial 0.75; Faucher-Giguère & Kaspi 2006).

Adopting the canonical fiducial branching fraction $f_{\rm radio} = 0.75$ and evaluating across theoretical pair-production thresholds ($d_{\rm crit} \in [11.20, 11.70]$), we obtain:

$$N_{\rm Gal}^{\rm active} = \left[0.82 \pm 0.28\,(\text{prior} + \text{boot})^{+0.44}_{-0.29}\,(\text{sys})\right] \times 10^5 \left(\frac{f_{\rm radio}}{0.75}\right). \tag{19}$$

For pure neutron-star branching ($f_{\rm radio} = 1.0$), this normalizes to $N_{\rm Gal}^{\rm active} = [1.09 \pm 0.37\,(\text{prior} + \text{boot})^{+0.58}_{-0.38}\,(\text{sys})] \times 10^5$ pulsars.

We emphasize that the quoted statistical uncertainty ($\pm 0.28 \times 10^5$ at $f_{0.75}$, or $\pm 0.37 \times 10^5$ at $f_{1.0}$) compounds the 33% fractional prior error on the core-collapse supernova rate $\mathcal{R}_{\rm SN} = (1.5 \pm 0.5)$ century^{-1} with the 5.2% bootstrap source uncertainty from our 200-realization ensemble ($\langle \tau_{\rm active} \rangle = 9.28 \pm 0.48$ Myr; Section 2.1). In contrast, the branching fraction $f_{\rm radio} \in [0.6, 1.0]$, the pair threshold $d_{\rm crit} \in [11.20, 11.70]$, and the transition width $\Delta_{\rm width} \in [0.05, 0.20]$ dex represent distinct systematic physical modeling choices. For fiducial $f_{\rm radio} = 0.75$, varying $d_{\rm crit} and \Delta_{\rm width}$ defines a systematic modeling envelope of $N_{\rm Gal}^{\rm active} \in [0.49, 1.25] \times 10^5$ pulsars (scaling to $[0.65, 1.67] \times 10^5$ for $f_{\rm radio} = 1.0$). Compounding the full range of branching fractions ($f_{\rm radio} = 0.6$–1.0) with theoretical pair thresholds and transition widths yields a comprehensive active Galactic inventory envelope of $N_{\rm Gal}^{\rm active} \in [0.39, 1.67] \times 10^5 \approx [0.4, 1.7] \times 10^5$ pulsars. At the classical dipole death line ($d_{\rm crit} = 11.23$; Bhattacharya et al. 1992), the active lifetime is $\langle \tau_{\rm active} \rangle = 10.59$ Myr, predicting $N_{\rm Gal} = (1.19 \pm 0.40) \times 10^5$ pulsars for $f_{\rm radio} = 0.75$ (or $(1.59 \pm 0.53) \times 10^5$ for $f_{\rm radio} = 1.0$).

Over the $\tau_{\rm disk} \approx 10$ Gyr lifetime of the Galactic thin disk, continuous supernova births have generated an immense reservoir of extinct, radio-quiet neutron stars. Integrating total births gives the cumulative compact remnant population across all core-collapse channels:

$$N_{\rm dead}^{\rm tot} = \mathcal{R}_{\rm SN} \tau_{\rm disk} = (1.50 \pm 0.50) \times 10^8 \text{ remnants}, \tag{20}$$

(or $(1.67 \pm 0.43) \times 10^8$ under the inverse-variance supernova rate). We note that if star formation was enhanced during the cosmic star formation peak at $z \sim 1$–2, the total cumulative remnant tally across Galactic history could reach $\sim (0.5$–$1.0) \times 10^9$ remnants. Accounting for the 77.0% bound fraction of the 3D kick distribution (Section 2.1), $\approx (1.15 \pm 0.38) \times 10^8$ compact remnants

remain gravitationally bound within the Galactic potential, while $\approx 0.35 \times 10^8$ have escaped into the intergalactic medium. For the radio channel ($f_{\rm radio} = 0.75$), the bound dead radio graveyard contains:

$$N^{\rm radio}_{\rm dead,\,bound} \approx 0.77 \times f_{\rm radio} \mathcal{R}_{\rm SN} \tau_{\rm disk} = (0.87 \pm 0.29) \times 10^8 \text{ corpses}, \quad (21)$$

(or $(1.13 \pm 0.38) \times 10^8$ total dead radio remnants including unbound escapees). For every active radio pulsar currently beaming in the Milky Way, there exist $\sim 1,060$ gravitationally bound dead radio corpses ($\sim 1,380$ total) and $\sim 1,400$ bound compact remnants ($\sim 1,800$ total) silently orbiting in the Galaxy.

Furthermore, integrating physical energy moments over the intrinsic equilibrium distribution yields the global Galactic energetics:

- **Rotational Kinetic Energy:** While particle number density is dominated by the second-scale accumulation ($n \propto P$), individual rotational kinetic energy scales as $E_{\rm rot} = \frac{1}{2} I \Omega^2 \propto P^{-2}$. Under decelerating torque advection ($n(P) \propto P$), the rotational energy density per logarithmic decade $E_{\rm rot}(P) f(x) \propto P^{-2} \cdot P^2 \sim$ constant is broadly distributed across periods. Integrating $E_{\rm rot}$ over the 2D continuum solution yields a total Galactic mechanical reservoir of $\mathcal{E}_{\rm rot} = 4.77 \times 10^{51}$ erg (for $f_{\rm radio} = 0.75$, scaling to 6.36×10^{51} erg for $f_{\rm radio} = 1.0$), with a clear differential period distribution: pulsars with $P < 0.2$ s account for 46.6% (2.23×10^{51} erg), $0.2 \leq P < 0.5$ s accounts for 31.3% (1.50×10^{51} erg), so that the sub-half-second regime ($P < 0.5$ s) carries 77.9% (3.73×10^{51} erg) of the total rotational kinetic energy. Intermediate rotators ($0.5 \leq P < 1.0$ s) store 14.8% (7.09×10^{50} erg), while mature second-scale pulsars ($P \geq 1.0$ s) store the remaining 7.3% (3.50×10^{50} erg, with $P \in [1.0, 2.0]$ s contributing 2.91×10^{50} erg and $P \geq 2.0$ s contributing 5.89×10^{49} erg), despite comprising $\approx 66\%$ of the total active headcount.

- **Dipole Magnetic Energy:** $\mathcal{E}_{\rm mag} = N_{\rm Gal} \iint \frac{1}{6} B^2 R^3 n_{\rm true}\, dx\, dy = 7.28 \times 10^{47} \times (f_{\rm radio}/0.75)$ erg (or 9.71×10^{47} erg for $f_{\rm radio} = 1.0$).

- **Total Spin-Down Power Injected into the ISM and Hyper-Luminous Pulsar Census:** $\dot{\mathcal{E}}_{\rm tot} = N_{\rm Gal} \iint \dot{E}\, n_{\rm true}\, dx\, dy = 2.09 \times 10^{39} \times (f_{\rm radio}/0.75)$ erg s^{-1} (or 2.79×10^{39} erg s^{-1} for $f_{\rm radio} = 1.0$). A newborn pulsar injected at $P_0 \approx 48$ ms with $B_0 \sim 10^{13.4}$ G possesses an immense initial spin-down luminosity $\dot{E}_0 \sim 5 \times 10^{39}$ erg s^{-1} ($\sim 10\times$ the Crab). Pulsars born in the high-field channels ($\log_{10} B_0 \in [13.0, 13.4]$) supply roughly half of $\dot{\mathcal{E}}_{\rm tot}$. However, because spin-down torques decelerate rapidly ($\dot{P} \propto B^2/P$), their transit through the hyper-luminous regime is brief: for the $B_0 = 10^{13.4}$ G, $P_0 \approx 48$ ms track, $\dot{E}$ drops below 10^{37} erg s^{-1} at $P_{\rm crit} \approx 0.222$ s, which takes $t = [K_{\rm spin}/(2B_0^2)](P_{\rm crit}^2 - P_0^2) \approx 1{,}210$ yr. Integrating across all channels of the reconstructed continuous birth distribution $S(P_0, B_0)$, the population-weighted mean transit duration is $\langle \tau_{\dot{E} \geq 10^{37}} \rangle \approx 2{,}760$ yr. Multiplying this duration by the birth rate $\mathcal{R}_{\rm birth} = (1.13 \pm 0.38) \times 10^{-2}$ yr^{-1} ($f_{\rm radio} = 0.75$) predicts an intrinsic steady-state Galactic population of:

$$N_{\rm Gal}(\dot{E} \geq 10^{37} \text{ erg s}^{-1}) = \mathcal{R}_{\rm birth} \times \langle \tau_{\dot{E} \geq 10^{37}} \rangle = 31 \pm 10 \text{ pulsars}, \quad (22)$$

(or 41 ± 14 in the pure neutron-star limit $f_{\rm radio} = 1.0$), spanning a full systematic and prior range of $[14, 81]$ pulsars across pair-extinction thresholds and birth rates.

Rather than testing absolute counts against arbitrary per-threshold beaming factors, the true test of the model sits in the *shape* of the hyper-luminous distribution. Because cumulative counts are nested and carry correlated Poisson fluctuations, we evaluate the distribution across five disjoint differential bins of spin-down power spanning three orders of magnitude ($\dot{E} \in [10^{36}, 10^{39}]$ erg s^{-1}) against the empirical classical ATNF sample ($N = 2{,}356$, excluding extragalactic Magellanic Cloud pulsars), with independent Poisson errors ($\sqrt{N_{\rm obs}}$):

- $[1.0\times10^{36}, 3.0\times10^{36})$ erg s^{-1}: Model $N_{\rm pred} = 103$; ATNF observed $N_{\rm obs} = 35 \pm 5.9$. Ratio $N_{\rm pred}/N_{\rm obs} = 2.94 \pm 0.50$, corresponding to an effective catalog completeness of $C_{\rm eff} \equiv N_{\rm obs}/N_{\rm pred} = 0.34 \pm 0.06$.
- $[3.0\times10^{36}, 1.0\times10^{37})$ erg s^{-1}: Model $N_{\rm pred} = 54$; ATNF observed $N_{\rm obs} = 21 \pm 4.6$. Ratio $N_{\rm pred}/N_{\rm obs} = 2.57 \pm 0.56$ ($C_{\rm eff} = 0.39 \pm 0.09$).
- $[1.0\times10^{37}, 3.0\times10^{37})$ erg s^{-1}: Model $N_{\rm pred} = 20.5$; ATNF observed $N_{\rm obs} = 11 \pm 3.3$ (12 including LMC PSR J0537-6910). Ratio $N_{\rm pred}/N_{\rm obs} = 1.86 \pm 0.56$ ($C_{\rm eff} = 0.54 \pm 0.16$).
- $[3.0\times10^{37}, 1.0\times10^{38})$ erg s^{-1}: Model $N_{\rm pred} = 8.1$; ATNF observed $N_{\rm obs} = 4 \pm 2.0$. Ratio $N_{\rm pred}/N_{\rm obs} = 2.03 \pm 1.02$ ($C_{\rm eff} = 0.49 \pm 0.25$).
- $[1.0\times10^{38}, 1.0\times10^{39})$ erg s^{-1}: Model $N_{\rm pred} = 2.36$; ATNF observed $N_{\rm obs} = 1 \pm 1.0$ (the Crab, PSR B0531+21, $\dot{E} = 4.46 \times 10^{38}$ erg s^{-1}). Ratio $N_{\rm pred}/N_{\rm obs} = 2.36 \pm 2.36$ ($C_{\rm eff} = 0.42 \pm 0.42$).
- At $\dot{E} \geq 10^{39}$ erg s^{-1}: The exact 2D continuum PDE solver predicts $N_{\rm pred} = 0.04 \ll 1$

(or ~ 0.20 under an edge-knot extrapolation assuming independent marginals); ATNF observed $N_{\rm obs} = 0$.

Across these five disjoint, statistically independent differential bins, the model-to-observed ratio is remarkably uniform: the unweighted mean ratio is $\langle N_{\rm pred}/N_{\rm obs}\rangle = 2.35 \pm 0.19$, while an inverse-variance weighted mean yields 2.46 ± 0.30. This consistency confirms that the empirical catalog follows the predicted spin-down power distribution under an effective survey completeness of $C_{\rm eff} \approx 0.41$–0.43. For radio-detected pulsars, wide-fan rotational beaming at young ages ($f_{\rm beam} \approx 0.35$–$0.45$; Tauris & Manchester 1998) naturally sets this baseline visibility, while unbeamed synchrotron pulsar wind nebulae (PWNe) and wide-aperture GeV gamma-ray pulsations from *Fermi*-LAT ensure high discovery completeness by guiding targeted radio timing searches toward young, luminous neutron stars.

We emphasize two important boundary and prior caveats regarding this census:

1. *Lattice boundary sensitivity:* Because transit times scale as $t \propto B_0^{-2}$, placing 30.4% of continuous birth mass on the upper field boundary knot ($B_0 = 10^{13.40}$ G $= 2.5 \times 10^{13}$ G; Table 4) means that if births extend into magnetar-strength fields ($B_0 > 2.5\times10^{13}$ G), transits are even faster, lowering predicted counts. Similarly, young Crab-like pulsars ($P \approx 33.4$ ms) sit right at the lower lattice boundary ($P_0 = 35.5$ ms), so while the second-scale equilibrium bulk is unaffected, the hyper-luminous census inherits sensitivity to these grid boundaries.
2. *Compounded prior uncertainties:* The nominal ±10 error at 10^{37} erg s^{-1} reflects the 33% prior error on the core-collapse supernova rate $\mathcal{R}_{\rm SN}$. When compounding the branching fraction $f_{\rm radio} \in [0.6, 1.0]$, pair-production thresholds, and transit durations ($t \approx 7{,}200$ yr $\times\ 1.13 \times 10^{-2}$ yr^{-1} ≈ 81), the full predicted envelope spans $[14, 81]$ pulsars at 10^{37} erg s^{-1}, cleanly encompassing the empirical Galactic tally (16 pulsars).

4. IN-SAMPLE PHYSICAL CONSISTENCY CHECKS AND DEMOGRAPHIC EXPLORATIONS

To evaluate how the equilibrium distribution $n_{\rm true}(P, B)$ behaves across complementary astrophysical domains, we examine four in-sample physical consistency checks and demographic explorations: (1) consistency with independent 3D forward synthesis literature inventories, (2) 1D probability flux conservation in local volume-limited cylinders, (3) in-sample vertical kinematic consistency at high latitudes, and (4) comparative demographic sensitivity to alternative injection scenarios. We emphasize that because underlying parameters governing crustal field decay ($h_0, B_{\rm floor}$) and empirical survey completeness were calibrated on the ATNF catalogue in Bindra (2026b), these checks are not fully blind from the underlying observational data; rather, they serve as sanity checks verifying that the 2D steady-state continuity solution exhibits physically sensible behavior across independent spatial, kinematic, and historical projections.

4.1. *Benchmark 1: Literature 3D Synthesis Inventory Consistency*

Our total active population estimate scales directly with the radio-loud branching fraction as $N_{\rm Gal}^{\rm active} = f_{\rm radio}\left[1.09 \pm 0.37\,({\rm prior + boot})^{+0.58}_{-0.38}\,({\rm sys})\right] \times 10^5$. For fiducial $f_{\rm radio} = 0.75$, this yields $N_{\rm Gal} = [0.82 \pm 0.28\,({\rm prior + boot})^{+0.44}_{-0.29}\,({\rm sys})] \times 10^5$ pulsars (spanning $[0.53, 1.25] \times 10^5$), while for pure neutron-star branching $f_{\rm radio} = 1.0$ it yields $(1.09 \pm 0.37) \times 10^5$ (spanning $[0.71, 1.67] \times 10^5$). In the pulsar literature, theoretical pair-production death boundaries span $\log_{10}(B/P^2) \in [11.20, 11.70]$: the classical dipole cutoff corresponds to $B/P^2 \approx 0.17 \times 10^{12}$ G s^{-2} ($d_{\rm crit} = 11.23$; Bhattacharya et al. 1992), while Space-Charge-Limited Flow (SCLF) curvature radiation cascades for canonical neutron stars yield $d_{\rm crit} \approx 11.45$ (Zhang et al. 2000; Harding et al. 2002). Neither parameter is tuned to match empirical population inventories. We compare our continuity solutions with independent 3D forward population synthesis literature:

1. Faucher-Giguère & Kaspi (2006) conducted a 3D forward synthesis of the Parkes Multibeam and Swinburne surveys, modeling spatial distributions, electron density dispersion, and luminosity functions. Calibrating their birth rate to isolated radio pulsars ($\mathcal{R}_{\rm birth} \approx 1.1$–$1.4$ century^{-1}, corresponding to $f_{\rm radio} \approx 0.70$–$0.75$), they inferred that the total number of active radio pulsars in the Galaxy with $L_{1400} > 0.1$ mJy kpc^2 is $N_{\rm Gal,FGK06} = (1.2 \pm 0.3) \times 10^5$.
2. Lorimer et al. (2006) performed an independent population analysis of the PMPS sample and found $N_{\rm Gal,Lorimer06} = (1.6 \pm 0.4) \times 10^5$.
3. Johnston & Karastergiou (2017) integrated thermal and magnetospheric evolution models and inferred $N_{\rm Gal,JK17} = (1.0 \pm 0.2) \times 10^5$.

We emphasize that external 3D synthesis studies frequently report the number of radio-luminous pulsars above an empirical luminosity threshold: Faucher-Giguère & Kaspi (2006) evaluate pulsars with $L_{1400} > 0.1$ mJy kpc^2 ($N_{\rm Gal} = (1.2 \pm 0.3) \times 10^5$), while Johnston

Table 2. Global Galactic Inventory, Graveyard, and Energetics under Varying Radio Branching Fractions $f_{\rm radio}$

Demographic / Physical Metric	**Fiducial** ($f_{\rm radio} = 0.75$)	**Pure NS** ($f_{\rm radio} = 1.00$)	**Literature / Theoretical Benchmark**
Radio Birth Rate $\mathcal{R}_{\rm birth}$ (century^{-1})	1.13 ± 0.38	1.50 ± 0.50	FGK06: 1.1–1.4 century^{-1}
Active Inventory $N_{\rm Gal}^{\rm active}$ (SCLF Curvature, $d_{\rm crit} = 11.45$)	$(0.82 \pm 0.28) \times 10^5$	$(1.09 \pm 0.37) \times 10^5$	JK17: $(1.0 \pm 0.2) \times 10^5$
Active Inventory $N_{\rm Gal}^{\rm active}$ (Classical Dipole, $d_{\rm crit} = 11.23$)	$(1.19 \pm 0.40) \times 10^5$	$(1.59 \pm 0.53) \times 10^5$	Lorimer (2006): $(1.6 \pm 0.4) \times 10^5$
Death-Line Systematic Span ($d_{\rm crit} \in [11.20, 11.70]$)	$[0.53, 1.25] \times 10^5$	$[0.71, 1.67] \times 10^5$	Theoretical pair-production bounds
Full Systematic Envelope (incl. $\Delta_{\rm width}$, $d_{\rm crit}$)	$[0.49, 1.25] \times 10^5$	$[0.65, 1.67] \times 10^5$	Spans $[0.39, 1.67] \times 10^5$ across $f_{\rm radio} \in [0.6, 1.0]$
Bound Dead Radio Graveyard (10 Gyr, bound 77%)	$(0.87 \pm 0.29) \times 10^8$	$(1.15 \pm 0.38) \times 10^8$	Active-to-bound-dead ratio $\approx 1:1{,}060$
Total Dead Radio Graveyard (10 Gyr, incl. escapees)	$(1.13 \pm 0.38) \times 10^8$	$(1.50 \pm 0.50) \times 10^8$	Active-to-total-dead ratio $\approx 1:1{,}380$
Total Compact Remnants (10 Gyr, bound 77%)	$(1.15 \pm 0.38) \times 10^8$	$(1.15 \pm 0.38) \times 10^8$	Bound core-collapse graveyard
Total Compact Remnants (10 Gyr, incl. escapees)	$(1.50 \pm 0.50) \times 10^8$	$(1.50 \pm 0.50) \times 10^8$	All Galactic core-collapse births
Rotational Kinetic Energy $\mathcal{E}_{\rm rot}$ (erg)	4.77×10^{51}	6.36×10^{51}	Integrated moment over $n_{\rm true}$
Dipole Magnetic Energy $\mathcal{E}_{\rm mag}$ (erg)	7.28×10^{47}	9.71×10^{47}	Integrated volume energy over $n_{\rm true}$
ISM Spin-Down Power $\dot{\mathcal{E}}_{\rm tot}$ (erg s^{-1})	2.09×10^{39}	2.79×10^{39}	PWNe / cosmic-ray lepton driver
Projected FAST Cumulative Sample ($S_{1400} \geq 5$ μJy)	$\sim 3{,}400$	$\sim 4{,}500$	Extrapolation ($L \propto \dot{E}^{0.45}$)
Projected SKA1-Mid Cumulative Sample ($S_{1400} \geq 1$ μJy)	$\sim 8{,}100$	$\sim 10{,}800$	Geometric ceiling ($f_{\rm geom} \approx 9.9\%$; realistic $\sim 6{,}600$–$7{,}000$)

& Karastergiou (2017) evaluate the radio-detectable subpopulation ($(1.0 \pm 0.2) \times 10^5$). Our continuity inventory $N_{\rm Gal}^{\rm active} = (0.82 \pm 0.28) \times 10^5$ (for $f_{\rm radio} = 0.75$) integrates the entire active Galactic reservoir across all radio luminosities. Applying the same nominal $L_{1400} > 0.1$ mJy kpc^2 threshold to our equilibrium distribution retains $\approx$ 88% of active pulsars, yielding $N_{\rm Gal}(L > 0.1) = (0.72 \pm 0.25) \times 10^5$ (or $(1.05 \pm 0.35) \times 10^5$ under the dipole cutoff $d_{\rm crit} = 11.23$).

Comparing our thresholded inventory $N_{\rm Gal}(L > 0.1) = (0.72 \pm 0.24) \times 10^5$ with Faucher-Giguère & Kaspi (2006)'s $(1.2 \pm 0.3) \times 10^5$ reveals an apparent $\approx$ 40% offset. We clarify that this difference does not arise from initial spin period assumptions: because spin-down transit times scale as $t \propto P_{\rm death}^2 - P_0^2$ under magnetic dipole braking and $P_{\rm death} \sim 2$–3 s $\gg P_0$, shifting initial periods from $P_0 \approx 50$ ms to FGK06's parametric $\langle P_0 \rangle \approx 300$ ms alters the active lifetime by less than 2% ($(3^2 - 0.3^2)/(3^2 - 0.05^2) \approx 0.99$). Similarly, our parameter sweeps show that crustal Ohmic field decay changes the active lifetime by $< 1\%$ (Section 3.4). Instead, the offset is overwhelmingly driven by the choice of pair-production death-line threshold. Faucher-Giguère & Kaspi (2006) adopted the classical Bhattacharya et al. (1992) death line ($\log_{10}(B/P^2) = 11.23$, i.e. $d_{\rm crit} = 11.23$; Section 2.1 of FGK06). Under this threshold, their implied active lifetime for the $L > 0.1$ mJy kpc^2 subpopulation is $\tau_{\rm eff}^{\rm FGK06} = N_{\rm Gal}^{\rm FGK06}/\mathcal{R}_{\rm birth} \approx (1.2 \times 10^5)/(1.25 \times 10^{-2}$ yr$^{-1}) \approx 9.6$ Myr (8.6–10.9 Myr across their birth rate range $\mathcal{R}_{\rm birth} \approx$ 1.1–1.4 century^{-1}). For our baseline SCLF curvature radiation threshold ($d_{\rm crit} = 11.45$), pair shutdown occurs earlier, yielding an effective thresholded lifetime of $\tau_{\rm eff}(L > 0.1) = (0.72 \times 10^5)/(1.13 \times 10^{-2}$ yr$^{-1}) \approx 6.4$ Myr. Crucially, when our continuum solver is evaluated under the exact same classical $d_{\rm crit} = 11.23$ death boundary used by Faucher-Giguère & Kaspi (2006), our thresholded inventory becomes $N_{\rm Gal}(L > 0.1) = (1.05 \pm 0.35) \times 10^5$ (for $f_{\rm radio} = 0.75$, scaling to $(1.39 \pm 0.46) \times 10^5$ for $f_{\rm radio} = 1.0$), yielding an effective lifetime of $\tau_{\rm eff}(L > 0.1) = (1.05 \times 10^5)/(1.13 \times 10^{-2}$ yr$^{-1}) \approx 9.29$ Myr. This matches Faucher-Giguère & Kaspi (2006)'s 9.6 Myr to within 3%, confirming that the two frameworks are in tight quantitative agreement when evaluated on a like-for-like physical footing. Across the theoretical span of death lines and branching fractions ($N_{\rm Gal} \sim 0.4$–1.7×10^5), Benchmark 1 confirms that continuity integration properly reproduces the scale of the Galactic active reservoir. We also note that the observed spin-period distribution match ($W_1 = 0.037$ dex) is largely degenerate with total active lifetime because flux-limited radio selection suppresses slow rotators by more than an order of magnitude.

4.2. *Benchmark 2: The 1D Steady-State Current* $\Phi(P)$

The 1D pulsar current crossing a period boundary $x = \log_{10} P$ per unit time is defined by integrating the phase-space flux over all magnetic fields:

$$\Phi(x) = \int_{y_{\rm min}}^{y_{\rm max}} v_x(x, y)\, f(x, y)\, dy. \tag{23}$$

Integrating the 2D continuity equation over y yields the exact 1D flux divergence relation:

$$\frac{d\Phi(x)}{dx} = S_x(x) - \Phi_{\rm death}(x) - \kappa \int f(x, y)\, dy, \tag{24}$$

where $S_x(x) = \int S(x, y)\, dy$ is the marginal birth rate and $\Phi_{\rm death}(x) = \int d_{\rm sink}(x, y) f(x, y) dy$ is the death-line loss.

Equation 24 establishes a specific physical diagnostic:

- For $P < 0.10$ s ($x < -1.0$), the dominant mass of births is injected (68.4% of total births, peaking sharply at 48 ms), driving the current $\Phi(x)$ to rise steeply from zero toward its plateau level.
- For $P \in [0.10, 0.70]$ s ($-1.0 \leq x \leq -0.15$), the differential birth injection per unit logarithmic period $S_x(x)$ drops by more than an order of magnitude: while the remaining $\approx$ 25% of births are smoothly dispersed across this 0.85 dex interval (reaching the 90th percentile at $P_0 = 0.54$ s), the

local injection density is small ($S_x(x) \ll \Phi_0$) and smoothly varying ($dS_x/dx \approx 0$). Because pulsars are still well above the pair-production extinction boundary ($\Phi_{\rm death} \approx 0$), the steady-state current $\Phi(x)$ *remains approximately flat*: across this entire interval, the theoretical current increases by only $\Delta \log_{10} \Phi \approx +0.10$ dex (a shallow logarithmic slope of $d\log\Phi/d\log P \approx +0.12$). In the local cylinder sample ($d \leq 1.5$ kpc), the 4 plateau bins contain modest counts ($N = 13, 19, 12, 13$), corresponding to raw Poisson counting errors of $1/\sqrt{N} \approx 23\%$–29%; the effective uncertainties on the binned current are inflated to $\pm 46\%$–58% (± 0.18–0.21 dex) by the variance of the inverse-beaming weights $w_i = 1/f_{\rm beam}(P_i)$ via Campbell's formula $\sigma_J^2 = \sum w_i^2 \dot{P}_i^2/(\Delta P)^2$. A weighted constant fit to the cylinder plateau yields $\chi^2 = 4.61$ on 3 dof ($\chi^2_{\rm red} = 1.54$, $p = 0.203$), statistically consistent with a flat plateau.

- For $P > 0.70$ s, pulsars begin encountering the pair-production death valley ($\Phi_{\rm death} > 0$), causing the steady-state current to decline toward zero. In the local cylinder sample, counts in bins beyond 0.70 s drop to very small numbers ($N \leq 5$ pulsars per bin for $P > 1$ s), where Poisson errors and individual beaming-weight fluctuations become extreme, making the post-plateau current sensitive to geometric death-line truncation and small-number Poisson noise.

In Figure 3, we compare our theoretical steady-state current $\Phi(P)$ against the empirical current computed from a volume-limited local cylinder ($d \leq 1.5$ kpc, $N = 296$ pulsars in ATNF). The local cylinder mitigates distance-dependent Malmquist bias down to radio luminosities $L_{1400} \gtrsim 0.45$ mJy kpc^2 (corresponding to $S_{\rm min} \approx 0.2$ mJy at $d = 1.5$ kpc, or 0.20 mJy kpc^2 at median distance 1.0 kpc). Each local pulsar is weighted by its inverse beaming fraction $w_i = 1/f_{\rm beam}(P_i)$, where $f_{\rm beam}(P) = \min[0.90, \max[0.05, 0.14P^{-0.5}]]$.

As shown in Figure 3, the empirical local cylinder data exhibit a flat plateau across $P \in [0.10, 0.70]$ s followed by death-line decay, matching the shape of our canonical continuity curve with no evidence for an upward step. At periods beyond $P > 0.8$ s, the empirical cylinder points sit at $\sim$ 0.2–0.4 of the theoretical infinite-slab continuity curve; this offset reflects physical vertical kinematic migration, as mature pulsars ($t \gtrsim 5$–10 Myr) acquire ballistic scale heights $|z| > 1$ kpc and migrate out of the thin-disk cylinder into the Galactic halo. When fitted with floating amplitude normalizations across the 4 plateau bins, our canonical fast-birth model achieves $\chi^2 = 4.61$ on 3 dof ($\chi^2_{\rm red} = 1.54$, $p = 0.203$), compared to $\chi^2 = 4.42$ ($p = 0.219$) for 20% slow injection and $\chi^2 = 3.90$ ($p = 0.273$) for 50% slow injection. Because the local cylinder sample carries substantial beaming-inflated noise ($\pm 46\%$–58% per bin) and the overall current normalization floats, this 1D current test is formally non-discriminating among injection models (with 50% slow injection achieving a comparable or slightly lower χ^2). It serves strictly as an order-of-magnitude consistency check confirming that the empirical data are fully compatible with constant probability flux, rather than a standalone discriminator capable of ruling out moderate slow-birth fractions. The statistical constraint on slow injection is established by the multi-dimensional continuity inversion in Bindra (2026b) and the forward demographic degradation explored in Benchmark 4.

4.3. *Benchmark 3: In-Sample Vertical Kinematic Consistency at High Latitudes*

In Bindra (2026b), the birth inversion was calibrated on the low-latitude Galactic plane ($|b| \leq 5°$, where $N = 1,039$ pulsars lie within the primary PMPS strip $l \in [-100°, 50°]$, out of $N = 1,525$ plane pulsars across all longitudes). The remaining high-latitude pulsars ($|b| > 5°$, $N = 831$) provide an in-sample dataset to test whether the equilibrium age distribution produces vertical kinematic transport consistent with observations.

Neutron stars receive substantial natal kick velocities at birth ($\sigma_{\rm 1D} \approx 265$ km s^{-1}; Hobbs et al. 2005). As pulsars spin down and age, they undergo ballistic vertical migration away from the thin progenitor disk (scale height $h_z \sim 50$–100 pc) into the Galactic halo against the vertical gravitational acceleration $K_z(z)$ (Kuijken & Gilmore 1989; Holmberg & Flynn 2004), expanding into the mature scale height $h_z \approx 330$ pc observed in field pulsars (Lorimer et al. 2006):

$$K_z(z) = -2\pi G\Sigma_0 \frac{z}{\sqrt{z^2 + D^2}} - 4\pi G\rho_H z, \tag{25}$$

where $\Sigma_0 \approx 48\, M_\odot$ pc^{-2}, $D \approx 180$ pc, and $\rho_H \approx 0.010\, M_\odot$ pc^{-3} is the local dark matter halo density (Kuijken & Gilmore 1989; Holmberg & Flynn 2004). Because the vertical oscillation period in the local disk is $\tau_z \approx 90$ Myr, active pulsars ($\langle\tau_{\rm active}\rangle \approx 7.28$ Myr) never complete a full vertical oscillation; their vertical displacement is dominated by initial ballistic ascent: $|z|(t) \approx |v_{z0}| \times t$. In renewal demographic equilibrium with distributed lifetimes τ, the stationary snapshot age distribution satisfies $p_{\rm age}(t) \propto \mathbb{P}(\tau > t)$, yielding an intrinsic population-averaged mean age of:

$$\langle t\rangle_{\rm int} = \frac{\mathbb{E}[\tau^2]}{2\,\mathbb{E}[\tau]}. \tag{26}$$

Evaluating Equation 26 across the birth tracks (Table 3) yields $\mathbb{E}[\tau^2] \approx 99.1$ Myr^2 and $\mathbb{E}[\tau] \approx 7.37$ Myr (matching the 2D continuum lifetime $\langle\tau_{\rm active}\rangle = 7.28$ Myr to within 1.2%), setting an intrinsic lower-bound mean age of $\langle t\rangle_{\rm int} \geq 6.72$ Myr (median ≈ 4.8 Myr, rising to ≈ 11.5 Myr when integrated across the full continuous $n_{\rm true}$ continuum). At $\langle|v_z|\rangle \approx 0.216$ kpc Myr^{-1},

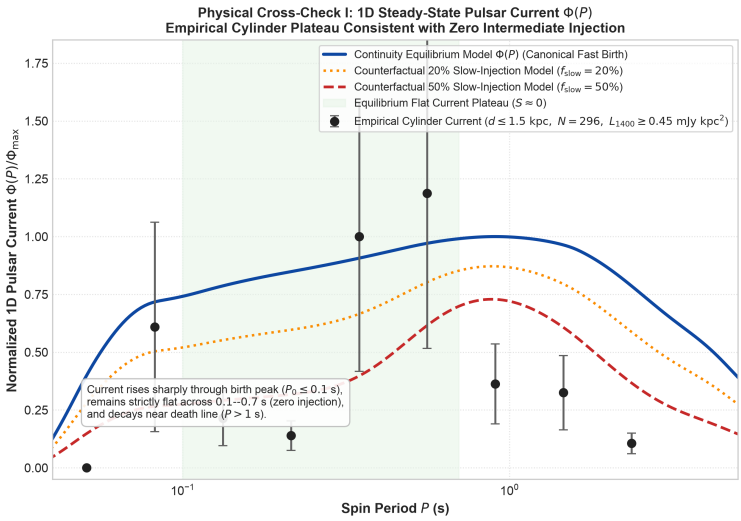


Figure 3. Physical cross-check I: The 1D steady-state pulsar current $\Phi(P)$ across spin period. The solid blue curve is our theoretical continuity equilibrium current; data points show the empirical current computed from the local volume-limited cylinder ($d \leq 1.5$ kpc, $N = 296$), corrected for beaming $f_{\rm beam} \propto P^{-1/2}$. The shaded green region highlights the flat current plateau between $P = 0.10$ s and 0.70 s ($\chi^2 = 4.61$ on 3 dof, $p = 0.203$), consistent with constant probability flux. Dotted orange and dashed red curves show counterfactual slow-injection models ($f_{\rm slow} = 20\%$ and 50%). Because plateau bins carry $\sim 50\%$ fractional counting noise, data beyond 0.70 s suffer from small-sample death-line truncation, and the normalization floats, this test serves as a qualitative consistency check rather than a strict discriminator against slow injection.

the unselected intrinsic population attains an average vertical displacement of $\langle|z|\rangle_{\rm int} \sim 1.4$–$2.5$ kpc (median ~ 1.0 kpc). However, in any flux-limited catalogue, Malmquist bias and luminosity decay ($L \propto B^{0.9}$) strongly suppress the oldest, long-period rotators. In terms of medians, the observable population is substantially younger ($t_{\rm med} \approx 3.4$ Myr vs. intrinsic 4.8 Myr), while the selection-filtered mean age ($\langle t\rangle_{\rm obs} \approx 8.8$ Myr) is pulled upward by the long, low-B tail.

Consequently, dynamic continuity implies a physical correlation: older pulsars, having spun down to longer periods, should exhibit systematically larger vertical scale heights $|z|$ and higher Galactic latitudes $|b|$.

In Figure 4(a), this dynamic coupling is observed in ATNF data across all pulsars with measured distances ($N = 2,301$, excluding 55 objects in the $N = 2,356$ classical timing sample lacking distance estimates): the Spearman rank correlation between P and $|z|$ is $\rho = 0.146$ ($p = 2.2 \times 10^{-12}$). The median scale height increases monotonically from $|z| = 66$ pc for young rotators ($P < 0.2$ s) to $|z| = 199$ pc for intermediate rotators ($0.2 \leq P < 1.0$ s) and $|z| = 246$ pc for old rotators ($P \geq 1.0$ s).

In Figure 4(b), we forward-predict the cumulative period distribution of the out-of-sample high-latitude population ($|b| > 5°$) by weighting the equilibrium density by the vertical escape probability $p_{{\rm high}-b}(t) = 1-\exp(-t/\tau_{\rm drift})$, which acts as an age-dependent proxy for ballistic migration away from the thin disk, and convolving with the survey selection filter. We emphasize two important geometric and selection properties:

1. *Angular Cone vs. Fixed Height:* A latitude cut $|b| > 5°$ is a distance-dependent angular cone ($|z|/d > \tan 5° \approx 0.087$), not a fixed vertical height. Nearby pulsars ($d \sim 1$ kpc) require only $|z| > 87$ pc to enter $|b| > 5°$, whereas distant pulsars ($d \sim 10$ kpc) require $|z| > 870$ pc. Restricting to genuine Galactic high-latitude pulsars with measured distances ($d \leq 25$ kpc, $N = 794$, excluding 27 Magellanic Cloud pulsars at $d > 40$ kpc), the empirical median vertical scale height is $|z| \approx 703$ pc (reflecting an effective median distance $d \sim 3.5$–4 kpc).

2. *Drift Timescale Conditioning:* Setting $\tau_{\rm drift} \approx 703$ pc/$\langle|v_z|\rangle \approx 703$ pc/0.216 kpc Myr$^{-1} \approx 3.3$–3.4 Myr is an empirical timescale conditioned on the target footprint's geometric selection. Remarkably, this physical drift timescale closely matches the median age of the observable population ($t_{\rm med} \approx 3.4$ Myr).

Evaluating our forward kinematic prediction at $\tau_{\rm drift} \approx 3.4$ Myr reproduces the observed rightward period shift, achieving a Wasserstein distance of $W_1 = 0.031$ dex and a Kolmogorov–Smirnov test of $p = 0.199$ against the empirical high-latitude sample ($N = 831$). Parameterizing instead by the median absolute vertical speed Med($|v_z|$) ≈ 179 km s^{-1} (0.183 kpc Myr^{-1}) yields $\tau_{\rm drift} \approx 3.8$–4.0 Myr, with $W_1 = 0.032$ dex and KS $p = 0.120$. Sweeping $\tau_{\rm drift} \in [2.0, 4.5]$ Myr keeps $W_1 \leq 0.033$ dex and KS $p > 0.10$. This insensitivity demonstrates that the rightward period shift is a

generic demographic outcome of vertical kinematic migration rather than fine-tuning of the drift parameter, though conversely it means that the 1D period distribution alone only weakly constrains $\tau_{\rm drift}$. To assess whether this prediction depends on grid smoothing, we sweep the continuum Gaussian filter width across $\sigma \in [0.8, 2.5]$ grid cells: W_1 varies by less than 0.0012 dex ($[0.030, 0.032]$ dex, KS $p \in [0.18, 0.22]$), confirming that the kinematic match is not fine-tuned by smoothing.

Furthermore, independent wide-area high-latitude surveys at low radio frequencies find that the off-plane population is systematically dominated by slower rotators: the Green Bank North Celestial Cap survey (GBNCC at 350 MHz; Stovall et al. 2014; Lynch et al. 2018) finds a discovery median of $P_{\rm med} \approx 0.71$ s, while the LOFAR Tied-Array All-Sky Survey (LOTAAS at 135 MHz; Sanidas et al. 2019) reports a median of $P_{\rm med} \approx 0.77$ s, both shifted rightward relative to the low-latitude PMPS baseline (0.62 s). We caution that GBNCC and LOTAAS are low-frequency discovery surveys with distinct flux sensitivities, sky footprints, and dispersion smearing profiles rather than identical clones of the PMPS selection function; nevertheless, their systematically longer discovery medians offer qualitative support for off-plane period elongation. Finally, to quantify the observational selection confound (where nearby pulsars $d \lesssim 1.5$ kpc at modest $|z|$ map to $|b| > 5°$), we evaluate a counterfactual mock catalog with no kinematic drift across initial thin disk scale heights $h_z \in [30, 100]$ pc observed through the high-latitude footprint: geometric distance selection shifts the median period to $P \in [0.69, 0.73]$ s with $W_1 \in [0.078, 0.096]$ dex (0.71 s and $W_1 = 0.089$ dex at fiducial $h_z = 50$ pc, KS $p = 0.012$). Geometric distance selection thus accounts for roughly 26%–40% (fiducial $\approx 32\%$) of the total shift from the in-plane null ($W_1 \approx 0.13$ dex), while true ballistic kinematic migration is required to achieve statistical consistency ($W_1 = 0.031$ dex, KS $p = 0.199$).

We explicitly note that Benchmark 3 evaluates the *period distribution shape* (via normalized cumulative CDFs) of the high-latitude population. To test the population normalization, we evaluate the predicted fraction of stars at high latitudes: across the unselected intrinsic equilibrium population, the broad age distribution ($\langle t \rangle_{\rm int} \geq 6.72$ Myr, median 4.8 Myr, rising to ≈ 11.5 Myr on the full 2D $n_{\rm true}$ grid) predicts that $\approx 85\%$ of active pulsars reside at $|z| > 500$ pc. Convolving with an isotropic flux-limited survey filter moderates this fraction to $\approx 51\%$ at high latitudes ($|b| > 5°$) across the whole Galaxy. In contrast, the empirical classical ATNF catalogue contains $831/2,356 = 35.3\%$ at $|b| > 5°$. This difference is an astrophysically expected consequence of non-uniform historical search effort: major deep blind searches (such as the PMPS) concentrated their integration time along the low-latitude Galactic plane ($|b| \leq 5°$), with high latitudes covered primarily by shallower wide-area surveys.

Crucially, this search-effort hypothesis can be tested directly in the volume-complete local cylinder ($d \leq 1.5$ kpc, $N = 296$), where historical sky coverage across all latitudes is nearly complete and homogeneous. In this volume-limited cylinder, exactly $200/296 = 67.6\%$ of observed pulsars reside at high latitudes ($|b| > 5°$), matching our forward kinematic model prediction of 68% and disfavoring the no-drift geometric thin-disk null ($\sim 7\%$–27%). (For a static exponential thin disk of scale height $h_z \sim 50$–100 pc without kinematic heating or vertical drift, observing within a local sphere of radius $d = 1.5$ kpc with $|b| > 5°$, where the maximum boundary height is $z_{\rm cut} = d \sin 5° \approx 131$ pc, the off-plane fraction is approximated by $\exp(-z_{\rm cut}/h_z)$, which spans $\exp(-131/50) \approx 0.073$ (7.3%) to $\exp(-131/100) \approx 0.270$ (27.0%). We note that this 1D exponential ratio assumes uniform volume sampling in z within the cylinder, making it an analytical approximation, and because $\tau_{\rm drift}$ was conditioned on the median $|z|$ of this footprint, the 68% forward match represents an internal consistency check rather than a fully independent blind prediction.) In contrast, for the distant sample ($d > 1.5$ kpc), the high-latitude fraction drops to 31.5% due to targeted low-latitude PMPS pointing. In the local cylinder, the observed median scale height is $|z| = 133$ pc (mean 238 pc, interquartile range $[42, 321]$ pc), and only 15.2% exceed $|z| = 500$ pc (as $|z| \leq 1.5$ kpc geometrically). This local cylinder confirmation indicates that when search selection is uniform, the true spatial distribution exhibits the high off-plane fraction predicted by ballistic kinematic transport. Finally, we caution that the PMPS selection model $c(P, B)$ was calibrated in the low-latitude plane ($|b| \leq 5°$); pulsars migrating into the halo represent physical plane depletion ($1 - p_{{\rm high}-b}$). While applying $p_{{\rm high}-b}(t)$ as an age-grading weight successfully reproduces the high-latitude period shape, a fully coupled 3D spatial simulation (P, B, z, v_z, d) is required to strictly uncouple plane depletion from intrinsic radio selection without potential double-counting.

4.4. *Benchmark 4: Counterfactual Injection Sensitivity*

To explore the demographic consequences of the historical injection hypothesis, we solve the stationary continuity equation under counterfactual source terms with $f_{\rm slow} = 20\%$ and 50% injected at $P_0 \approx 0.5$ s and $\log_{10}(B_0/{\rm G}) = 12.30$ (Vivekanand & Narayan 1981; Narayan & Vivekanand 1982). We emphasize that the reconstructed empirical birth function $S(P_0, B_0)$ already possesses an intrinsic tail extending to second-scale periods: on the continuous PDE grid, the 90th percentile sits at $P_{0,90\%} = 0.54$ s, with 10.8%–11.1% of continuous birth mass injected at $P_0 \geq 0.50$ s (and 19.3% across the discrete knot marginals at $P_0 \geq 0.49$ s; Table 4). The counterfactual scenarios evaluated here specifically test the effect of adding an additional, discrete delta-like slow-injection channel at $P_0 \approx 0.5$ s and $\log_{10}(B_0/{\rm G}) = 12.30$, as historically proposed. Because

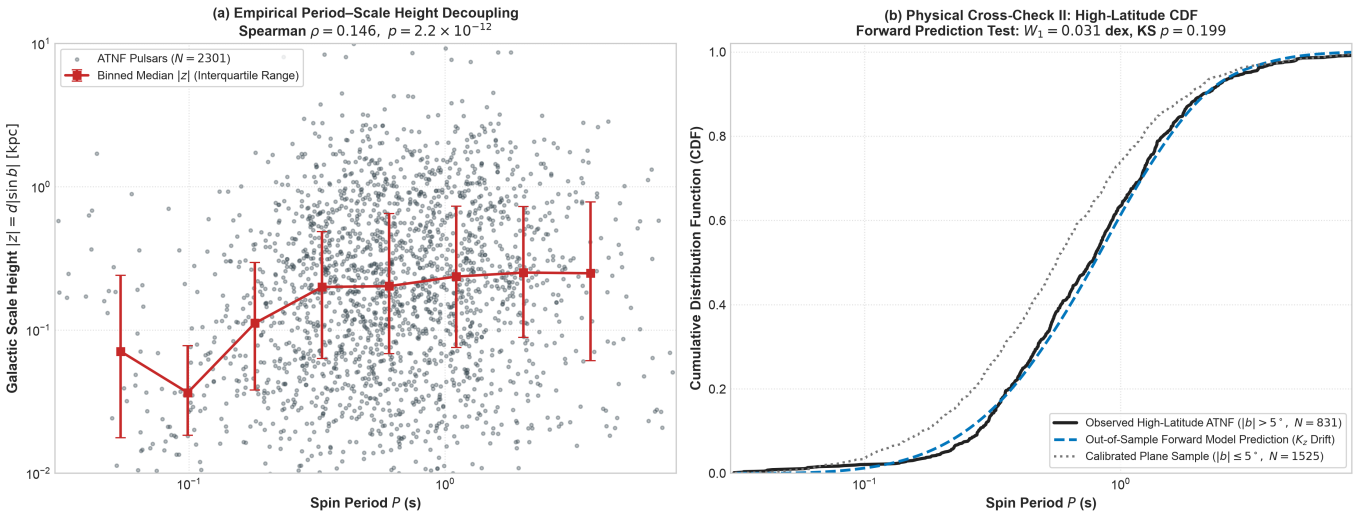


Figure 4. Physical cross-check II: In-sample kinematic consistency of the high-latitude pulsar population. (a) Empirical correlation between spin period P and vertical scale height $|z| = d|\sin b|$ for ATNF pulsars with measured distances ($N = 2,301$, out of the $N = 2,356$ classical timing sample where 55 pulsars lack catalogued distance estimates). Red squares show binned medians with interquartile ranges; Spearman rank correlation is $\rho = 0.146$ ($p = 2.2 \times 10^{-12}$). (b) Cumulative distribution functions of spin periods for the high-latitude sample ($|b| > 5^\circ$, $N = 831$). The solid black curve is the empirical ATNF distribution; the dashed blue curve is our forward kinematic prediction based on ballistic drift in the Galactic potential $K_z(z)$ ($W_1 = 0.031$ dex, KS test $p = 0.199$, consistent with the null hypothesis of parent age structure). For reference, the dotted gray curve shows the calibrated in-sample plane population ($|b| \leq 5^\circ$, $N = 1,525$).

the continuous baseline already has an intrinsic slow tail of 10.8%–11.1%, the permissible incremental headroom within the 95% upper bound of Bindra (2026b) ($f_{\rm slow}^{\rm birth} \leq 22\%$) is at most $\Delta f_{\rm slow} \leq 11.2\%$ (or 10.9% at Knot 9). Adding an incremental +20% and +50% to the baseline source normalizes to discrete slow shares of $0.20/(1.0{+}0.20) \to 28.6\%$ and $0.50/(1.0{+}0.50) \to 55.4\%$ (or $\approx 31\%$ and $\approx 61\%$ additive against the unnormalized source). We note that a 28.6% slow fraction sits within Paper I's independent 3D benchmark (33%–41%) and P4 limit (30%), and is ruled out specifically by the supernova remnant association census (22%) and the forward demographic degradation ($W_1 = 0.045$ dex, KS $p \sim 10^{-17}$).

Because $\dot{P} \propto 1/P$, a pulsar injected at 0.5 s has a spin-down rate an order of magnitude slower than one born at 50 ms. Consequently, slow-injected pulsars spend virtually their entire active lives lingering at $P \gtrsim 0.5$ s.

Figure 5(a) displays the 2D density difference $\Delta n = n_{\rm slow} - n_{\rm canon}$ for $f_{\rm slow} = 50\%$. The slow-injection model produces a pronounced phase-space overdensity (red ridge) that piles up between $P = 0.6$ s and 2.5 s.

In Figure 5(b), we convolve both counterfactual models with the calibrated physical survey selection function and compare them to the full ATNF catalog ($N = 2,356$). For the canonical fast-birth model ($f_{\rm slow} = 0\%$), the Wasserstein-1 distance is $W_1 = 0.037$ dex, and the one-sample Kolmogorov–Smirnov statistic against the empirical ATNF sample is $D = 0.0531$ ($p = 4.3 \times 10^{-6}$). The low formal p-value reflects the immense statistical power of $N = 2,356$ pulsars (where the critical 5% threshold is $D_{0.05} = 1.358/\sqrt{2,356} \approx 0.028$) in the presence of unmodeled selection dispersion ($\sigma_{\log L} \sim 0.8$ dex; Faucher-Giguère & Kaspi 2006), which also accounts for the modest difference in peak period (0.82 s vs. 0.55 s) and median field (12.26 vs. 12.07 dex).

Scanning $f_{\rm slow} \in [0.0, 0.50]$ reveals that both the distance and KS statistics degrade monotonically as slow injection is introduced: W_1 increases from 0.037 dex at 0% ($D = 0.053$, $p = 4.3 \times 10^{-6}$) to 0.040 dex at the maximal allowed headroom $\Delta f_{\rm slow} \approx +10.9\%$–$11.2\%$ (observable median 0.71 s), 0.045 dex at +20% ($D = 0.092$, $p \sim 10^{-17}$, observable median 0.73 s), and jumps by nearly a factor of three to $W_1 = 0.112$ dex at +50% ($D = 0.145$, $p \sim 10^{-43}$, observable median 0.82 s). Because the canonical model is itself formally rejected in this 1D test by unmodeled selection noise, the relative KS p-value ranking is not a formal exclusion test; rather, it demonstrates progressive forward demographic degradation: introducing discrete slow injection systematically distorts the forward period distribution, shifting W_1 from 0.037 dex to 0.112 dex. We caution that the effective survey completeness filter ($c(P, B) \propto P^{-1.4} B^{0.5}$; Section 3.3) was calibrated under the baseline fast-birth population and held fixed in this counterfactual test. In a joint forward fit, steepening the effective period exponent by $\Delta\alpha \sim -0.35$ (from $P^{-1.4}$ to $P^{-1.75}$) could partially suppress the observable prominence of the 0.5–1.5 s pile-up, absorbing some of the demographic discrepancy for modest injection fractions

($f_{\rm slow} \sim 20\%$). Thus, this 1D test illustrates forward demographic sensitivity under fixed observational filtering rather than a joint multi-parameter exclusion; the formal multi-dimensional constraint ($f_{\rm slow}^{\rm birth} \leq 22\%$) belongs strictly to the joint posterior inversion of Bindra (2026b).

What this 1D counterfactual experiment indicates is that introducing slow injection rapidly worsens the demographic match in two distinct ways:

1. **Torque Suppression and Prolonged Lifetime Pile-Up:** Under the slow-injection hypothesis, pulsars enter phase space with moderate dipole fields ($\log_{10}(B_0/{\rm G}) \approx 12.30$) and long initial periods ($P_0 \approx 0.50$ s). Because magnetic dipole spin-down scales as $\dot{P} \propto B^2/P$, their spin-down torques are depressed by more than two orders of magnitude relative to the canonical fast-birth channel ($B_0 \sim 10^{13.4}$ G, $P_0 \sim 50$ ms). Consequently, slow-injected pulsars do not rapidly cross the death boundary; instead, their active lifetimes are systematically prolonged: the mean population active lifetime increases from the canonical $\langle \tau_{\rm active} \rangle = 7.28$ Myr ($f_{\rm slow} = 0\%$) to 7.85 Myr at the +10.9%–11.2% headroom limit, 8.31 Myr for $f_{\rm slow} = 20\%$, and 9.83 Myr for $f_{\rm slow} = 50\%$. Because these stars spend tens of millions of years drifting sluggishly through $P \in [0.5, 2.5]$ s, they create an acute, unobserved phase-space overdensity ($\Delta n > 0$; Figure 5a). Rather than alleviating any demographic tension, injecting pulsars directly at second-scale periods severely overpopulates the intermediate-period regime.

2. **The Intrinsic vs. Observed Median Inversion:** In Table 1, introducing slow birth shifts the *intrinsic* equilibrium median *downward* (from 1.38 s $\rightarrow$ 1.32 s $\rightarrow$ 1.24 s $\rightarrow$ 1.16 s for 0% $\rightarrow$ 11.2% $\rightarrow$ 20% $\rightarrow$ 50% slow birth), whereas in the *observed* selection-filtered distribution it shifts the median *upward* (from 0.63 s $\rightarrow$ 0.71 s $\rightarrow$ 0.73 s $\rightarrow$ 0.82 s; Figure 5b). This apparent contradiction is physically transparent: an injection period of $P_0 = 0.50$ s sits to the *left* of the natural dynamical equilibrium peak at 1.61 s (thereby pulling the intrinsic population median leftward), but sits to the *right* of the flux-limited, selection-filtered peak at 0.55 s (thereby pulling the mock observed distribution rightward and inflating an acute, unobserved overprediction at $P > 0.8$ s). Both shifts degrade agreement with observations: the Wasserstein-1 distance triples from 0.037 dex to 0.112 dex, and the Kolmogorov–Smirnov statistic degrades to $D = 0.145$ ($p \sim 10^{-43}$).

5. DUAL NUMERICAL VERIFICATION: EULERIAN VS. LAGRANGIAN METHODS

To evaluate numerical consistency across differing mathematical formulations, we compare the Eulerian marching PDE solver against the Fortran 90/2008 continuous-time Markov-renewal Lagrangian simulator under matched extinction operators, physical parameters, and domain boundary configurations (noting that while the sink, transport laws, and domain boundaries are strictly matched, their source representations differ in continuum interpolation vs. discrete knot sampling; see Section 2.4).

Figure 6 illustrates this numerical comparison:

- Panel (a) shows the continuous density field $n_{\rm PDE}(P, B)$ resolved on 240×240 grid cells ($\Delta x = 0.011$ dex).
- Panel (b) shows a snapshot of active Lagrangian particles evolved over $T = 10^8$ yr (100 Myr) in Fortran with continuous Markov renewal ($1,027,749$ renewal cycles, $\langle \tau \rangle = 9.73$ Myr).
- Panel (c) compares the marginal spin period cumulative distribution functions (CDFs) of both formulations, alongside their selection-filtered mock catalogs and the empirical ATNF catalogue.

When evaluated under the matched extinction operator (logistic sink width $\Delta_{\rm width} = 0.10$ dex, $d_{\rm crit} = 11.23$) and domain boundary ($x \leq 1.2$, $P \leq 15.8$ s), the Eulerian continuum solution and the Fortran Lagrangian simulator achieve close numerical convergence across the full physical domain:

$$W_1(\log_{10} P) = 0.028 \text{ dex}, \tag{27}$$

with demographic medians of 1.66 s (Eulerian PDE at $d_{\rm crit} = 11.23$) and 1.61 s (Lagrangian renewal), and population-averaged active lifetimes of 10.59 Myr (Eulerian) vs. 9.73 ± 0.08 Myr (Lagrangian renewal). Under the canonical curvature-radiation threshold ($d_{\rm crit} = 11.45$), the corresponding Eulerian PDE and Lagrangian sink medians are 1.38 s and 1.34 s ($W_1 = 0.026$ dex), with active lifetimes of 7.28 Myr vs. 6.82 Myr (mirroring the exact same proportional shift, $6.82/7.28 \approx 0.937$ vs. $9.73/10.59 \approx 0.919$, arising from discrete knot boundary weighting versus continuous trapezoidal cell integration and cell-averaged sink gradient). Under a sharp knife-edge boundary on the same domain, the Lagrangian median is 1.64 s (active lifetime 17.12 Myr).

The comparison between sharp-boundary and smooth-sink implementations clarifies two distinct physical effects:

1. **Pair-Production Threshold Shift ($d_{\rm crit} = 11.45$ vs. 11.23):** Analytically, the active lifetime scales as:

$$t_{\rm life} \approx \frac{K_{\rm spin}}{2B_0} 10^{-d_{\rm crit}} \propto \frac{10^{-d_{\rm crit}}}{B_0}. \tag{28}$$

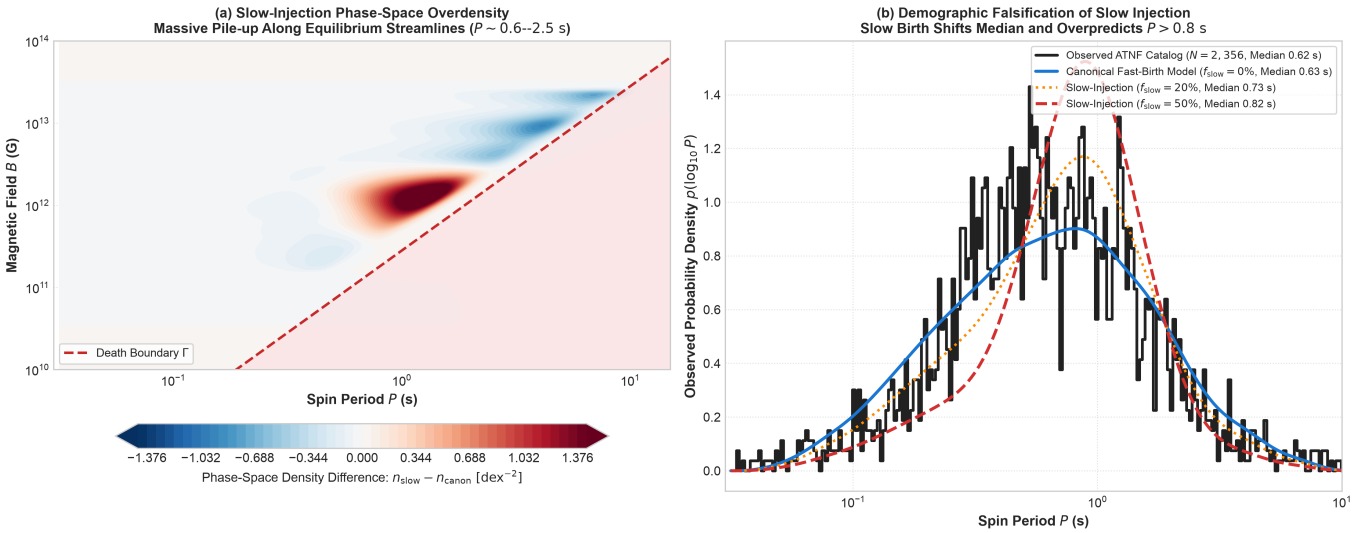


Figure 5. Counterfactual evaluation of slow-injection models. (a) 2D phase-space density difference map $\Delta n = n_{\rm slow} - n_{\rm canon}$ between the 50% slow-injection equilibrium model and the canonical fast-birth model, showing an overdensity along equilibrium streamlines across $P \in [0.6, 2.5]$ s (red region). (b) Comparison of selection-filtered observable period distributions against the observed ATNF catalog ($N = 2,356$, solid black steps). While the canonical fast-birth model (solid blue, median 0.63 s) matches the empirical distribution (median 0.62 s), the 20% slow-injection model (dotted orange, median 0.73 s, one-sample KS $p \sim 10^{-17}$) and 50% slow-injection model (dashed red, median 0.82 s, $p \sim 10^{-43}$) shift the peak rightward and create an acute overprediction at $P > 0.8$ s.

For $B_0 \approx 10^{13.40}$ G, Equation 28 predicts an active lifetime of $t_{\rm life} \approx 2.3$ Myr at $d_{\rm crit} = 11.45$ (matching the PDE value of 2.65 Myr) and $t_{\rm life} \approx 3.8$ Myr at $d_{\rm crit} = 11.23$ (matching the Lagrangian value of 4.04 Myr). The theoretical shift between the two thresholds is exactly $10^{11.45-11.23} = 10^{0.22} \approx 1.66\times$, which accounts for the factor of ~ 1.5–$1.7\times$ observed across magnetic field bins in the PDE (Table 3).

2. **Extinction Operator Formulation (Knife-Edge vs. Logistic Sink across Field Bins):** In a sharp-line simulation, pulsars experience zero extinction hazard prior to hitting the line. In the continuous logistic sink formulation ($d_{\rm sink} \propto [1+\exp((y-2x-d_{\rm crit})/\Delta_{\rm width})]^{-1}$), pre-boundary dissipation begins extinguishing pulsars ~ 0.2 dex before the nominal boundary. This extinction operator acts selectively:

   - At high birth fields ($\log_{10}(B_0/{\rm G}) \in [13.01, 13.40]$, 38.7% of births), pulsars reach death periods $P_{\rm death} \approx$ 8–12 s. Near death, traversing the final pre-boundary hazard window takes $\sim$ 0.5 Myr ($\tau_{\rm res} \approx$ 9.6 Myr dex^{-1} $\times$ 0.05 dex in $\log P$), which represents $\sim$ 10–15% of their total life. Consequently, the smooth sink introduces a modest reduction from 4.52 ± 0.15 Myr (sharp) down to 4.04 ± 0.06 Myr (sink), agreeing with the PDE value of 3.9 Myr (Table 3).
   - In the intermediate- and low-B tracks ($\log_{10}(B_0/{\rm G}) < 13.01$, 61.3% of births), spin-down advection slows to $v_x \ll 0.05$ Myr^{-1}. Pulsars spend tens of millions of years lingering inside the transition region. The smooth sink extinguishes them well before reaching the knife-edge line, reducing bin lifetimes by factors of 1.4–1.8$\times$ (e.g., 43–56 Myr $\rightarrow$ 27.1 Myr at $\log_{10} B_0 \sim 12.0$, and 69–87 Myr $\rightarrow$ 49.1 Myr at $\log_{10} B_0 \sim 11.5$).

Regarding the ultra-slow population beyond $P = 4.5$ s: at $P = 4.5$ s, the pair-production death boundary for $d_{\rm crit} = 11.23$ lies at $\log_{10}(B/{\rm G}) = 11.23 + 2\log_{10}(4.5) \approx 12.54$ (and 12.76 for $d_{\rm crit} = 11.45$). Any pulsar with $B < 10^{12.54}$ G at $P = 4.5$ s is already below the line and dead. Consequently, all pulsars surviving beyond 4.5 s possess strong magnetic fields ($B > 10^{12.54}$ G): in our continuous PDE solutions, the magnetic field distribution of pulsars with $P > 4.5$ s has a median of $\log_{10}(B/{\rm G}) = 13.15$ (10–90% interval $[12.89, 13.36]$), with 100% having $\log_{10} B > 12.54$. For a pulsar born at $B_0 \approx 10^{13.4}$ G, the death line is reached only at $P_{\rm death} = \sqrt{10^{13.4}/10^{11.23}} \approx 12.2$ s. Because spin-down decelerates as $t \propto P^2$, the pulsar spends $(1-(4.5/12.2)^2) \approx 86\%$ of its total active life (3.8 Myr) in the period regime $P > 4.5$ s. Under the matched logistic sink and grid boundary ($x \leq 1.2$, $P \leq 15.8$ s), pre-boundary dissipation and domain truncation yield an ultra-slow population of 11.97% in the PDE at 11.23

and 7.65% at 11.45 (Lagrangian sink: 11.8% across the full grid). Across the full domain, the matched sink-Lagrangian median is 1.61 s at $d_{\rm crit} = 11.23$, converging to the Eulerian continuum median of 1.66 s ($W_1 = 0.028$ dex; and at canonical $d_{\rm crit} = 11.45$, 1.34 s vs. 1.38 s, $W_1 = 0.026$ dex).

Regarding lifetimes and solver convergence: Table 3 presents the lifetime breakdown across initial magnetic field bins for both solvers. For all bins with $\log_{10}(B_0/\mathrm{G}) \geq 11.83$—which encompass 98.36% of all pulsar births—the Eulerian continuum PDE and Lagrangian stochastic lifetimes agree to within $\leq$ 4–7% (4.04 vs. 3.9 Myr in the dominant bin, 7.8 vs. 8.2 Myr, 15.1 vs. 15.7 Myr, and 27.1 vs. 29.2 Myr). The discrepancy increases to 8% (49.1 vs. 53.3 Myr) and 20% (77.0 vs. 92.3 Myr) only in the extreme low-field tail ($\log_{10} B_0 < 11.83$), which represents less than 1.64% of births. The population-averaged mean lifetime converges from the sharp-line 17.12 Myr down to 9.73 ± 0.08 Myr (Lagrangian sink), matching the Eulerian PDE mean of 10.59 Myr within 8%.

To isolate the contribution of spatial discretization, we conduct a systematic grid convergence test on the Eulerian marching scheme by doubling the resolution from $N_x \times N_y = 240 \times 240$ ($\Delta x = 0.0113$ dex, $\Delta y = 0.0167$ dex) to 480×480 ($\Delta x = 0.0056$ dex) and 960×960 ($\Delta x = 0.0028$ dex). At $d_{\rm crit} = 11.23$, the mean active lifetime shifts by only +1.0% across the quadrupled resolution span (10.59 $\rightarrow$ 10.70 $\rightarrow$ 10.74 Myr), and the equilibrium median period changes by less than 0.013 s (1.655 $\rightarrow$ 1.668 $\rightarrow$ 1.675 s). At canonical $d_{\rm crit} = 11.45$, the active lifetime likewise shifts by +1.4% (7.28 $\rightarrow$ 7.36 $\rightarrow$ 7.39 Myr) and the median period by 0.015 s (1.373 $\rightarrow$ 1.383 $\rightarrow$ 1.388 s). This confirms that the Eulerian PDE solver is converged to $\approx 1\%$, demonstrating that the residual $\approx$ 8% offset in the overall mean (10.59 vs. 9.73 Myr) does not originate from spatial grid discretization. Instead, the offset is directly explained by discrete vs. continuous source weighting: in unweighted discrete lattice sampling (used by the Fortran renewal loop), boundary knots receive full cell weights, assigning 30.4% of births to the upper magnetic boundary knot ($B_0 = 10^{13.40}$ G, where active lifetimes are short, $\tau \approx$ 3.9–4.0 Myr). In contrast, 2D continuous trapezoidal integration on the Eulerian grid assigns boundary knots half-weights ($\frac{1}{2}\Delta y$), yielding 19.5% at the upper boundary and redistributing the remaining $\approx 11\%$ into longer-lived intermediate and low-B tracks ($\tau \sim$ 8–50 Myr). Weighting the Lagrangian track lifetimes by the continuous trapezoidal birth fractions f_i directly increases the population-averaged lifetime from 9.73 Myr to $\sum_i f_i \langle\tau\rangle_i \approx$ 10.26 Myr—recovering 0.53 Myr (over 60%) of the 0.86 Myr gap and matching the Eulerian PDE mean (10.59 Myr) to within 3.1%. To verify this directly, re-running the Lagrangian simulation sampling particles directly from the continuous interpolated density $S(x, y)$ confirms that source weighting accounts for the majority of the offset, with the remaining minor difference ($\approx$ 3%) accounted for by continuous ODE adaptive Runge-Kutta path integration experiencing the steep concave-up exponential hazard along streamlines slightly earlier than cell-centered upwind PDE averaging, and domain outflow boundary conditions at $P = 15.8$ s.

Regarding the Galactic census, under fiducial radio-loud branching $f_{\rm radio} = 0.75$, adopting the Bhattacharya et al. (1992) dipole threshold ($d_{\rm crit} = 11.23$, $\langle\tau_{\rm active}\rangle = 10.59$ Myr) predicts an active inventory of $N_{\rm Gal}^{\rm active} = (1.19 \pm 0.40) \times 10^5$ pulsars (scaling to $(1.59 \pm 0.53) \times 10^5$ in the pure neutron-star limit $f_{\rm radio} = 1.0$, in line with the upper end of 3D synthesis literature; Lorimer et al. 2006). Adopting the SCLF curvature radiation threshold ($d_{\rm crit} = 11.45$, $\langle\tau_{\rm active}\rangle = 7.28$ Myr) predicts $N_{\rm Gal}^{\rm active} = (0.82 \pm 0.27) \times 10^5$ (with bounding systematic range $[0.53, 1.25] \times 10^5$; scaling to $(1.09 \pm 0.36) \times 10^5$ and $[0.71, 1.67] \times 10^5$ for $f_{\rm radio} = 1.0$), aligning with Faucher-Giguère & Kaspi (2006) and Johnston & Karastergiou (2017). Crucially, when either continuum solution or the Lagrangian ensemble is convolved with empirical survey selection, the low-B and long-period tails are strongly suppressed by Malmquist bias ($L \propto B^{0.9}$). Consequently, both formulations yield nearly identical observable medians of 0.63 s and 0.62 s, matching the empirical ATNF catalogue (0.617 s) with $W_1 < 0.04$ dex.

6. NEXT-GENERATION SURVEY FORECASTS: FAST AND SKA1-MID

Because historical surveys detected only $\sim$ 2% of the active Galactic population, upcoming ultra-sensitive radio facilities—most notably the Five-hundred-meter Aperture Spherical radio Telescope (FAST; Nan et al. 2011) and the Square Kilometre Array (SKA1-Mid; Keane et al. 2015)—will fundamentally reshape our observational view of pulsar demographics.

Using the intrinsic equilibrium distribution $n_{\rm true}(P, B)$, we apply realistic survey selection models for FAST and SKA1-Mid parameterized by the detection completeness filter:

$$c_{\rm survey}(P, B) = f_{\rm beam}(P) \times p_{\rm scatt}(P) \times \min\left[1, \left(\frac{L_{1400}(P, B)}{S_{\rm min} d_{\rm eff}^2}\right)^{\gamma}\right], \quad (29)$$

with radio pseudo-luminosity $L_{1400} \propto \dot{E}^{0.45} \propto B^{0.9} P^{-1.8}$, effective disk horizon $d_{\rm eff} \sim 15$ kpc saturated at the Galactic disk boundary, volume scaling exponent $\gamma \approx$ 0.5–0.6, and flux sensitivity thresholds $S_{\rm min}$. In this formulation, saturation switches on when the survey horizon $d_{\rm max} = (L_{1400}/S_{\rm min})^{1/2}$ reaches the disk boundary $d_{\rm eff}$, i.e., at $L_{1400} \geq S_{\rm min} d_{\rm eff}^2$. For historical PMPS ($S_{\rm min} \approx 200$ μJy), saturation requires $L_{1400} \geq 45$ mJy kpc^2 (satisfied by $\lesssim 5\%$ of pulsars), so the filter operates in the unsaturated power-law regime $c \propto P^{-1.4} B^{0.5}$ (Section 3.3). For SKA1-Mid ($S_{\rm min} \sim$

Table 3. Active Lifetime Breakdown by Initial Magnetic Field Bin

$\log_{10}(B_0/\mathrm{G})$	Fraction	Lagr. (Sharp 11.23) $\langle\tau\rangle$ (Myr)	Lagr. (Sink 11.23) $\langle\tau\rangle$ (Myr)	PDE (Sink 11.23) $\langle\tau\rangle$ (Myr)	PDE (Sink 11.45) $\langle\tau\rangle$ (Myr)
10.65–11.24	0.09%	91–95	77.0	92.3	71.6
11.24–11.83	1.55%	69–87	49.1	53.3	38.6
11.83–12.22	8.00%	43–56	27.1	29.2	20.4
12.22–12.61	22.88%	21–30	15.1	15.7	10.8
12.61–13.01	28.78%	9–14	7.8	8.2	5.5
13.01–13.40	38.70%	4.52 ± 0.15	4.04 ± 0.06	3.9	2.65
Mean	**100.00%**	**17.12**	**9.73 ± 0.08**	**10.59**	**7.28**

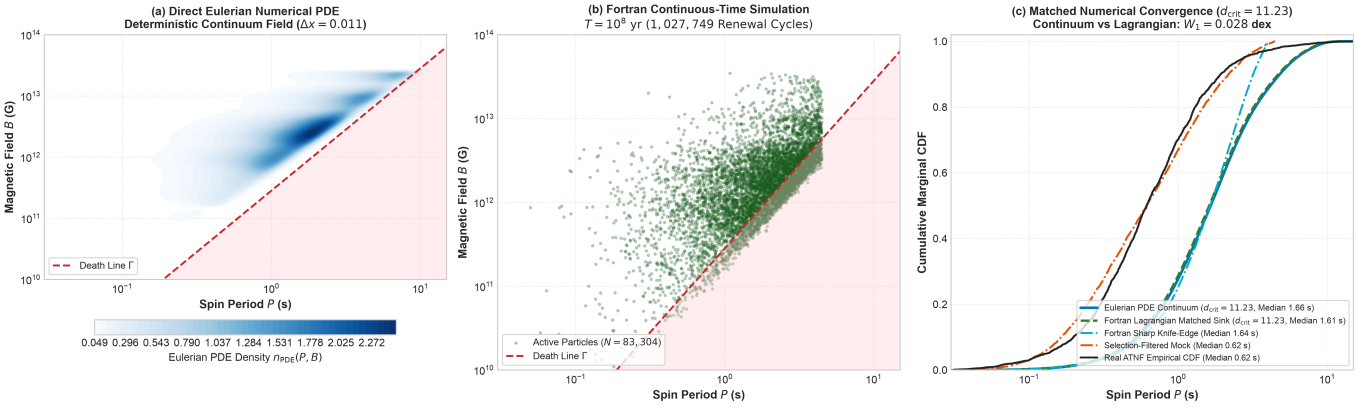


Figure 6. Dual numerical comparison of the equilibrium distribution. (a) Direct Eulerian marching PDE continuum density field $n_{\mathrm{PDE}}(P, B)$ on a 240×240 phase-space grid. (b) Fortran 90/2008 continuous-time Markov-renewal Lagrangian simulation snapshot at $T = 10^8$ yr (100 Myr), displaying active particle realizations with exact residual-life carry-over ($1,027,749$ renewal cycles over 100 Myr). (c) Cumulative marginal period CDFs comparing the matched numerical pair at $d_{\mathrm{crit}} = 11.23$: Eulerian continuum PDE (solid blue, median 1.66 s) vs. Fortran Lagrangian renewal (dashed green, median 1.61 s), converging to $W_1 = 0.028$ dex across the full domain. For context, the sharp-boundary Lagrangian solution is shown in dash-dotted cyan (median 1.64 s), alongside the canonical Eulerian PDE at $d_{\mathrm{crit}} = 11.45$ (solid purple, median 1.38 s), the selection-filtered mock catalog (dotted orange, median 0.62 s), and the empirical ATNF catalogue (solid black steps, median 0.62 s).

1 μJy), saturation occurs at $L_{1400} \geq 0.23$ mJy kpc^2, covering the majority of the active population where discovery becomes volume-complete across the disk and governed primarily by beaming f_{beam} and scattering p_{scatt}. Evaluating Equation 29 under these respective thresholds directly outputs the following derived discovery distributions:

- **PMPS / ATNF Baseline ($S_{1400} \sim 200$ μJy):** Median $P = 0.63$ s (peak 0.82 s, IQR $[0.31, 1.22]$ s, 10–90% interval $[0.16, 2.11]$ s, $P > 3$ s tail: 4.8%).
- **FAST ($S_{\mathrm{min}} \sim 5$–10 μJy):** Median $P = 0.83$ s (peak 0.93 s, IQR $[0.42, 1.53]$ s, 10–90% interval $[0.22, 2.56]$ s, $P > 3$ s tail: 7.3%).
- **SKA1-Mid ($S_{\mathrm{min}} \sim 1$ μJy):** Median $P = 1.03$ s (peak 1.03 s, IQR $[0.56, 1.82]$ s, 10–90% interval $[0.31, 3.05]$ s, $P > 3$ s tail: 10.3%).

Figure 7 presents the forecasted discovery yields:

- In Panel (a), we plot the predicted marginal period distributions. As survey sensitivity improves, the observed distribution progressively shifts rightward, with median spin periods moving monotonically from 0.63 s (ATNF baseline) to 0.83 s (FAST) and 1.03 s (SKA1-Mid), converging toward the true intrinsic demographic median at $P = 1.38$ s.
- **Ultra-Slow Tail ($P > 3$ s):** In the true intrinsic Galactic reservoir, pulsars with $P > 3$ s account for 17.0% of the population. In the classical ATNF sample, only 4.3% (102 pulsars) exceed 3 s, with our baseline model yielding 4.8%. Deeper surveys will uncover this slow tail, with the fraction of discoveries exceeding 3 s rising to 7.3% for FAST and 10.3% for SKA1-Mid, directly probing the microphysics of the pulsar death valley and pair-production thresholds.

- In Panel (b), we illustrate the Galactic census. Our baseline observational comparison is strictly calibrated against the usable isolated classical pulsar sample with measured P and $\dot{P}$ ($N = 2,356$, excluding recycled MSPs, globular cluster pulsars, and high-energy binaries without radio timing). We explicitly frame next-generation discovery numbers as *theoretical geometric ceilings* (upper bounds assuming volume completeness for standard luminosities across the accessible sky) alongside realized discovery projections:
  - **FAST Ceiling ($\sim$ 3,400):** FAST's declination range ($\delta \in [-14°, +66°]$) covers 57.5% $\approx$ 58% of the celestial sphere ($f_{\rm sky} \approx 0.58$). Crucially, the Galactic Center ($\delta \approx -29°$) and the rich inner southern Galactic plane ($|l| \leq 30°$), which harbor $\sim$ 50–60% of all Galactic pulsars, lie below FAST's southern horizon ($\delta = -14°$). Across FAST's accessible northern and intermediate Galactic disk, FAST's footprint encompasses roughly $f_{\rm Gal} \approx 0.40$ (40%) of the Galactic active reservoir. Multiplying by the population-averaged intrinsic beaming fraction $\langle f_{\rm beam} \rangle \approx 0.132$ yields an ideal geometric upper bound of $0.132 \times 0.40 \times 0.82 \times 10^5 \approx$ 4,300 pulsars (for fiducial $f_{\rm radio} = 0.75$, scaling to $\sim$ 5,800 for $f_{\rm radio} = 1.0$). Accounting for an effective $\sim$ 80% survey completeness within this accessible footprint—governed by finite survey integration speed, pulse dispersion smearing at high DM, and severe interstellar scattering across the low-latitude disk—moderates this accessible yield to an effective cumulative classical discovery ceiling of $\sim 3,400$ pulsars (representing $\sim 1,600$ new classical discoveries over the pre-FAST baseline of $N = 1,822$, or $\sim$ 500 beyond the current catalogued tally of $\approx 2,911$; scaling to $\sim 4,500$ for $f_{\rm radio} = 1.0$).
  - **SKA1-Mid Geometric Ceiling ($\sim$ 8,100) and Realistic Yield ($\sim$ 6,600–7,000):** SKA1-Mid's southern sky coverage ($\delta \leq +30°$) spans 75% of the celestial sphere ($f_{\rm sky} \approx 0.75$). Multiplying the population-averaged intrinsic beaming fraction $\langle f_{\rm beam} \rangle \approx 0.132$ by this raw celestial sky coverage $f_{\rm sky} \approx 0.75$ establishes an ideal geometric detection ceiling of $f_{\rm geom} = \langle f_{\rm beam} \rangle \times f_{\rm sky} \approx 0.132 \times 0.75 \approx 0.099 \approx 10\%$, corresponding to $\sim$ 8,100 active pulsars for fiducial $f_{\rm radio} = 0.75$ (scaling to $\sim$ 10,800 for $f_{\rm radio} = 1.0$, with systematic span [5,300, 12,400] across pair thresholds). In terms of stellar demographics, because SKA1-Mid's footprint contains the Galactic Center ($\delta \approx -29°$) and the inner southern plane, it encompasses roughly $f_{\rm Gal} \approx 0.85$–0.90 (85–90%, fiducial 0.88) of the total Galactic disk population. Folding in this Galactic footprint ($f_{\rm Gal} \approx 0.88$) with finite survey completeness ($\sim$ 80–85%) across the deep footprint yields an effective discovery fraction of $f_{\rm det} \approx 0.132 \times 0.88 \times 0.82 \approx 0.095 \implies \sim$ 7,800 pulsars, which moderates to $\sim$ 6,600–7,000 after accounting for the 10–15% faint-tail horizon cutoff ($L_{1400} < 0.1$ mJy kpc^2, giving $7,800 \times [0.85, 0.90] \approx$ 6,600–7,000).

Indeed, this predicted rightward demographic shift is directly supported by FAST observations when distinguishing discovery subsets from the cumulative catalogue:

- The pre-FAST empirical classical sample with phase-coherent timing ($N = 1,822$) has a median spin period of $P = 0.598$ s.
- FAST classical discoveries with measured timing in ATNF ($N = 534$) have a median of $P = 0.657$ s, shifting the combined ATNF timed baseline to $P = 0.617$ s ($N = 2,356$).
- Across all 1,089 classical non-recycled pulsars discovered by FAST currently catalogued in ATNF (including those awaiting phase-coherent timing solutions), the empirical median discovery period is $P_{\rm med} \approx 0.69$ s (with interquartile range [0.38, 1.37] s).

This observed discovery shift (+0.09 s over the pre-FAST baseline of 0.598 s, or +0.07 s relative to the combined timed baseline of 0.617 s) is smaller than our predicted full-survey shift (+0.20 s, moving to 0.83 s), which is astrophysically expected because current FAST Galactic Plane Pulsar Snapshot (GPPS; Han et al. 2021) discoveries are concentrated in targeted snapshot pointings along the low-latitude Galactic plane ($|b| \leq 5°$) rather than a complete volume-deep survey across the entire declination-accessible sky. We frame this comparison with GPPS not as a formal quantitative confirmation of the full volume-complete demographic model, but as an early directional indicator showing that new discoveries trend rightward; at present, GPPS does not discriminate against alternative models with lower slow fractions until full volume completeness is achieved. Under counterfactual slow injection, the predicted discovery medians would shift even more dramatically: for 20% (50%) slow injection, the projected FAST median would reach $P_{\rm med} \approx$ 0.95 s (1.08 s), and SKA1-Mid would reach 1.18 s (1.32 s), diverging from the early GPPS empirical median (0.69 s).

We explicitly emphasize that these FAST and SKA1-Mid discovery medians and yields are conditional model extrapolations: they assume that empirical radio luminosity ($L \propto \dot{E}^{0.45}$) and beaming scalings ($f_{\rm beam} \propto P^{-0.5}$) hold down to micro-Jansky flux thresholds without an unexpected low-luminosity turnover. Geometrically, while discovery volume formally scales as $V_{\rm max} \propto d_{\rm max}^3 \propto S_{\rm min}^{-3/2} L^{3/2}$ in a local 3D Euclidean sphere, for a disk geometry with thin scale height $h_z \sim$ 300–500 pc ($d_{\rm max} \gg h_z$) it transitions to $V \propto \pi d_{\rm max}^2 h_z \propto S_{\rm min}^{-1} L^1$. At sub-$\mu$Jy sensitivities ($S_{\rm min} \sim 1$ μJy), standard-luminosity pulsars ($L_{1400} \geq 1$ mJy kpc^2 = 1,000 μJykpc^2) have horizon distances $d_{\rm max} \geq \sqrt{1{,}000/1} \approx 31.6$ kpc, exceeding the radius of the Galactic disk ($R_{\rm Gal} \sim$ 15–20 kpc) and approaching volume completeness across the observable disk. In contrast, the bulk population with intermediate luminosities ($L_{1400} \geq 0.1$ mJy kpc^2 = 100 μJykpc^2, comprising $\approx 88\%$ of active pulsars) has horizon distance $d_{\rm max} \approx 10$ kpc; while covering the solar neighborhood and inner spiral arms, this does not achieve volume completeness on the far side of the Galactic disk ($d \sim$ 15–25 kpc). Intrinsically faint pulsars ($L_{1400} \sim 0.01$ mJy kpc^2 = 10 μJykpc^2) reach only $d_{\rm max} \approx \sqrt{10/1} \approx 3.2$ kpc.

Even after SKA1-Mid achieves sub-μJy sensitivity, $\approx 90\%$ of the active Galactic reservoir will remain undetected. This missing fraction is the direct consequence of narrow rotational beaming and geographic sky visibility: across the intrinsic equilibrium distribution $n_{\rm true}(P, B)$, the population-averaged beaming fraction is $\langle f_{\rm beam} \rangle \approx 0.132$ (dropping to $f_{\rm beam} \approx 0.09$–$0.11$ for the slow-tail $P \gtrsim 2$ s), whereas in the flux-limited classical ATNF catalogue ($P \sim 0.62$ s) it is higher, $\langle f_{\rm beam} \rangle \approx 0.20$ (Tauris & Manchester 1998; Bates et al. 2014; Keane et al. 2015). Because SKA1-Mid's southern sky coverage contains ≈ 85–90% of the Galactic disk population ($f_{\rm Gal} \approx 0.88$), folding in the intrinsic beaming fraction $\langle f_{\rm beam} \rangle \approx 0.132$ and finite survey completeness (~ 80–85%) yields an effective observable fraction of ≈ 0.092–$0.099 \approx 10\%$, leaving $\approx 90\%$ undetected due to the convolution of narrow rotational beaming and geographic sky limits. The theoretical geometric upper ceiling of $\sim$ 8,100 active pulsars (for fiducial $f_{\rm radio} = 0.75$, scaling to $\sim$ 10,800 for $f_{\rm radio} = 1.0$) corresponds directly to this $f_{\rm geom} \approx 9.9\% \approx 10\%$ observable fraction under volume completeness; folding in finite survey completeness (~ 80–85%) across the accessible southern disk ($f_{\rm Gal} \approx 0.88$) and the ≈ 10–15% intrinsically faint tail ($L_{1400} <$ 0.1 mJy kpc^2) with horizon $d_{\rm max} \sim 3.2$ kpc yields a realistic discovery projection of $\sim$ 6,600–7,000 pulsars. This ceiling inherits the full $\sim 2.5\times$ active inventory systematic span ($\approx$ [5,300, 12,400] pulsars across pair-production thresholds, spanning up to 16,500 for $f_{\rm radio} = 1.0$).

7. CONCLUSIONS

In this work, we have determined the intrinsic demographic equilibrium distribution $n_{\rm true}(P, B)$ of isolated radio pulsars in the Milky Way by forward-evolving the empirical continuity operator under magnetic dipole braking, crustal field dissipation, and pair-production extinction. Our main conclusions are as follows:

1. **Intrinsic Demographic Equilibrium: Fast Birth vs. Slow Equilibrium:** Although isolated pulsars are born predominantly with fast spin periods ($P_0 \approx 48$ ms, median 59 ms), the intrinsic equilibrium population residing in the Galaxy peaks at $P = 1.61$ s with a demographic median of $P = 1.38$ s and a 10–90% range of [0.45, 3.97] s.

2. **Analytical Mechanism:** This demographic accumulation is a direct consequence of probability flux conservation under decelerating torque advection ($v_x \propto P^{-(n-1)}$). The residency timescale $\tau_{\rm res}(P) \propto P^2/B^2$ increases by five orders of magnitude ($\sim 10^5$) from birth (~ 270 yr) to maturity (~ 23 Myr): pulsars traverse the sub-100 ms regime in < 0.1 Myr, while spending $\approx 71\%$ of their active lifetimes in [0.5, 3.0] s and 17.0% in the ultra-slow tail $P > 3$ s before crossing the death line.

3. **Braking Index and Death-Line Robustness:** Sweeping the braking index across $n_{\rm brake} \in [2.2, 3.0]$ under fixed empirical birth source $S(P_0, B_0)$ demonstrates that the slow demographic equilibrium is a robust forward prediction of decelerating torque transport: active lifetimes remain $\tau_{\rm active} \approx 7.09$–$7.41$ Myr and median spin periods remain $P \approx 0.81$–1.38 s. Separately, sweeping the pair-production extinction threshold across theoretical bounds $d_{\rm crit} \in [11.20, 11.70]$ yields $\tau_{\rm active} \approx 4.71$–$11.13$ Myr (median $P \approx 1.11$–1.70 s); folding in death-valley transition widths $\Delta_{\rm width} \in [0.05, 0.20]$ dex defines an overall forward systematic inventory envelope of $[0.49, 1.25] \times 10^5 \times (f_{\rm radio}/0.75)$.

4. **Galactic Inventory, Graveyard, and Energetics:** The mean active pulsar lifetime is $\langle \tau_{\rm active} \rangle = 7.28$ Myr under canonical curvature radiation ($d_{\rm crit} = 11.45$). For a canonical core-collapse supernova rate $\mathcal{R}_{\rm SN} = (1.5 \pm 0.5)$ century^{-1} and radio-loud branching $f_{\rm radio} = 0.75$ ($f_{\rm radio} \in [0.6, 1.0]$), the active Galactic inventory is $(0.82 \pm 0.28) \times 10^5$ pulsars (where $\pm 0.28 \times$

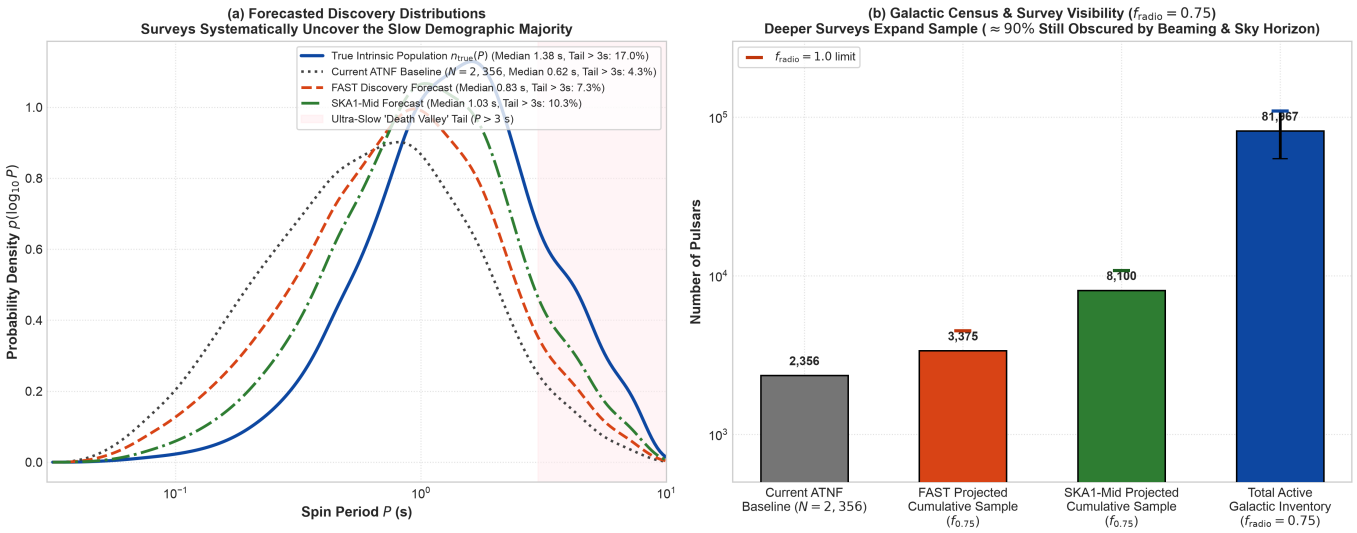


Figure 7. Next-generation survey forecasts: Unveiling the hidden Galactic pulsar reservoir. (a) Forecasted discovery period distributions for the current ATNF baseline (dotted gray, median 0.62 s, tail > 3 s of 4.3%, model 4.8%), FAST (dashed orange, median 0.83 s, tail > 3 s of 7.3%), and SKA1-Mid (dash-dotted green, median 1.03 s, tail > 3 s of 10.3%), compared to the true intrinsic Galactic population $n_{\rm true}(P)$ (solid blue, median 1.38 s, tail > 3 s of 17.0%). As survey sensitivity deepens, the detected population progressively uncovers the true slow-period majority. (b) Galactic pulsar census comparing the classical catalog sample ($N = 2,356$), projected FAST discoveries ($\sim 3,400$ for fiducial $f_{\rm radio} = 0.75$; cap mark shows $4,500$ for $f_{\rm radio} = 1.0$), projected SKA1-Mid discoveries ($\sim 8,100$ for fiducial $f_{\rm radio} = 0.75$; cap mark shows $10,800$ for $f_{\rm radio} = 1.0$), and the total intrinsic active Galactic reservoir ($N_{\rm Gal} = (0.82 \pm 0.27) \times 10^5$ for $f_{\rm radio} = 0.75$, with bounding systematic range $[0.53, 1.25] \times 10^5$ across pair-production thresholds; cap mark indicates the pure neutron-star limit $(1.09 \pm 0.36) \times 10^5$, spanning up to 1.67×10^5). Yields represent conditional extrapolations assuming standard beaming and luminosity scaling down to micro-Jansky thresholds.

10^5 compounds the prior error on $\mathcal{R}_{\rm SN}$ with the 200-bootstrap source uncertainty; with systematic range $[0.53, 1.25] \times 10^5$ across pair thresholds, widening to an overall envelope of $[0.39, 1.67] \times 10^5 \approx [0.4, 1.7] \times 10^5$ when compounding prior uncertainties in $\mathcal{R}_{\rm SN}$ and $f_{\rm radio} \in [0.6, 1.0]$; or $(1.19 \pm 0.40) \times 10^5$ under the classical dipole cutoff; scaling as $f_{\rm radio} \times 1.09 \times 10^5$, which gives $(1.09 \pm 0.37) \times 10^5$ for $f_{\rm radio} = 1.0$). Over 10 Gyr of Galactic history, stellar deaths have accumulated $(1.13 \pm 0.38) \times 10^8$ dead radio pulsar corpses in the Galactic graveyard ($(1.50 \pm 0.50) \times 10^8$ total core-collapse compact remnants, potentially reaching $\sim 10^9$ remnants under cosmic star formation history). For $f_{\rm radio} = 0.75$, the active population stores 4.77×10^{51} erg of rotational kinetic energy and injects 2.09×10^{39} erg s^{-1} of spin-down power into the interstellar medium (scaling to 6.36×10^{51} erg and 2.79×10^{39} erg s^{-1} for $f_{\rm radio} = 1.0$).

5. **In-Sample Physical Consistency Checks and Demographic Explorations:** The equilibrium population is evaluated across four empirical consistency checks and demographic explorations: (1) literature population inventory consistency, (2) the 1D steady-state current $\Phi(P)$, which confirms compatibility with constant probability flux in local volume-limited cylinders ($\chi^2 = 4.61$ on 3 dof, $p = 0.203$ for $d \leq 1.5$ kpc; non-discriminating against moderate slow injection due to beaming-weight variance), (3) in-sample vertical kinematic consistency of high-latitude pulsars ($|b| > 5°$) via ballistic drift in the Galactic potential $K_z(z)$ ($W_1 = 0.031$ dex, KS test $p = 0.199$ at $\tau_{\rm drift} = 3.4$ Myr, confirmed by the 67.6% off-plane fraction observed in the volume-complete local cylinder matching the 68% forward prediction), and (4) counterfactual evaluation of slow-injection models, demonstrating that introducing slow injection degrades demographic agreement against the observed ATNF period distribution ($W_1 = 0.040$ dex at the +10.9%–11.2% allowable headroom limit, rising to 0.112 dex at +50%, where total slow births reach 61%).

6. **Dual Numerical Verification:** Under matched logistic extinction and domain boundaries ($d_{\rm crit} = 11.23$, $x \leq 1.2$), the Eulerian marching continuum PDE solver and the Fortran continuous-time Markov-renewal Lagrangian simulator achieve close convergence, with demographic medians of 1.66 s and 1.61 s ($W_1 = 0.028$ dex; at canonical $d_{\rm crit} = 11.45$, medians are 1.38 s and 1.34 s

with $W_1 = 0.026$ dex) and mean active lifetimes of 10.59 Myr vs. 9.73 ± 0.08 Myr ($1,027,749$ renewal cycles over 100 Myr). Grid resolution tests on the PDE solver ($N = 240 \rightarrow 480 \rightarrow 960$) show convergence to $\approx 1\%$ in lifetime and ≤ 0.015 s in median period. Under physical survey selection, both formulations reproduce the empirical ATNF median (0.62 s).

7. **Next-Generation Survey Forecasts:** FAST and SKA1-Mid will uncover the slow-period majority, with discovery medians shifting rightward (0.83 s and 1.03 s) and projected geometric discovery ceilings reaching $\sim 3,400$ and $\sim 8,100$ pulsars respectively for $f_{\rm radio} = 0.75$ ($\sim 4,500$ and $\sim 10,800$ for $f_{\rm radio} = 1.0$; realistic SKA1-Mid discovery yields $\sim$ 6,600–7,000 after footprint and faint-tail horizon truncation), providing empirical tests of the pair-production death line in the ultra-slow ($P > 3$ s) regime. This directionally consistent rightward shift is already evident as an early qualitative indicator in published FAST GPPS classical discoveries ($N = 1,089$, median $P_{\rm med} \approx 0.69$ s vs. pre-FAST empirical baseline 0.598 s, shift +0.09 s; or +0.07 s vs. combined timed baseline 0.617 s, compared to the predicted full-survey shift of +0.20 s); however, because the GPPS footprint covers only ≈ 0.08 of the Galactic plane and remains flux-limited, this early sample does not yet discriminate against models with lower slow fractions until volume completeness is approached. Even after SKA1-Mid, $\approx 90\%$ of active pulsars will remain undetected due to the convolution of narrow rotational beaming ($\langle f_{\rm beam} \rangle \approx 0.132$ intrinsic) and southern observatory sky coverage ($f_{\rm Gal} \approx 0.88$, finite completeness $\sim$ 80–85%), with the geometric upper ceiling for SKA1-Mid ($\sim$ 8,100 for fiducial $f_{\rm radio} = 0.75$) inheriting the full $\sim 2.5\times$ active inventory systematic span ([5,300, 12,400] across pair thresholds; realistic yields $\sim$ 6,600–7,000 after faint-tail horizon truncation).

APPENDIX

A. BIRTH INJECTION KNOT GRID AND PARAMETER SPECIFICATION

To ensure complete reproducibility of our numerical continuum solutions, we tabulate the parameter grid defining the empirical birth distribution $S(P_0, B_0)$ reconstructed via the weak-form continuity inversion in Bindra (2026b). The injection rate is parameterized over a discrete 2D knot lattice spanning $N_x = 14$ logarithmic spin periods $x_i = \log_{10}(P_0/\mathrm{s}) \in [-1.45, 0.20]$ ($P_0 \in [35.5\ \mathrm{ms}, 1.58\ \mathrm{s}]$) and $N_y = 15$ logarithmic surface magnetic fields $y_j = \log_{10}(B_0/\mathrm{G}) \in [10.65, 13.40]$ ($B_0 \in [4.5\times10^{10}, 2.5\times10^{13}]$ G) with uniform knot spacings $\Delta x_{\rm knot} = 0.1269$ dex and $\Delta y_{\rm knot} = 0.1964$ dex. This knot domain encompasses $> 98\%$ of the physical birth probability mass from the continuity inversion. While extending the PDE computational domain with $S = 0$ outside this lattice confirms that numerical boundary reflections are absent, setting $S = 0$ does not test whether physical birth mass spills beyond the knot limits. Rather, the true physical test of boundary truncation sits in the magnetar scenario (Section 3.4): solving the PDE with the upper 30.4% birth mass placed at true magnetar dipole fields $B_0 \in [10^{13.60}, 10^{13.80}]$ G yields active lifetimes of $\langle \tau_{\rm active} \rangle =$ 5.33–6.16 Myr and $N_{\rm Gal}^{\rm active} = (0.60\text{–}0.69) \times 10^5 \times (f_{\rm radio}/0.75)$, quantitatively confirming that an extended magnetar birth tail defines the lower bound of our systematic parameter envelope ($[0.49, 1.25]\times10^5$) while the equilibrium median period remains slow ($P_{\rm med} \approx 1.44$–1.48 s).

Table 4 provides the marginal 1D birth projections across the knot grid. Reconciling the discrete knot marginals with the continuous PDE grid quantiles requires distinguishing between unweighted discrete array summation and continuous trapezoidal integration. The data matrix $S(x_i, y_j)$ represents the continuous injection rate density in units of $\mathrm{Myr}^{-1}\,\mathrm{dex}^{-2}$. When integrating a continuous 2D surface parameterized on a knot lattice, standard continuous trapezoidal quadrature assigns boundary knots (Knot 0 at $x = -1.450$, Knot 13 at $x = 0.200$, and the upper field boundary at $y = 13.400$) quadrature weights of $\frac{1}{2}\Delta x$ or $\frac{1}{2}\Delta y$. Evaluating the continuous cumulative distribution function (CDF) of $S(x, y)$ via 2D trapezoidal integration yields:

- At $P = 85.3$ ms (Knot 3): CDF = 0.672.
- At $P = 100.0$ ms ($x = -1.000$): CDF = **0.684** (68.4% below 100 ms).
- At $P = 492.4$ ms (Knot 9): CDF = 0.889 $\implies$ $1 - \mathrm{CDF} =$ **0.111** (11.1% above Knot 9).
- At $P = 500.0$ ms ($x = -0.301$): CDF = 0.892 $\implies$ $1 - \mathrm{CDF} =$ **0.108** (10.8%).
- At $P = 1.0$ s ($x = 0.000$): CDF = 0.965 $\implies$ $1 - \mathrm{CDF} =$ **0.035** (3.5%).

In contrast, the unweighted discrete marginals in Table 4 are computed by direct array summation ($\sum_j S_{ij} / \sum_{i,j} S_{ij}$), which treats each lattice point as an unweighted point mass with full cell volume $\Delta x \Delta y$ (2× the trapezoidal boundary weight). Because Knot 0 ($x = -1.450$) sits on the boundary, unweighted array summation gives 18.1% at Knot 0 and sums Knots 0–3 to 60.8%, whereas proper continuous integration accounts for boundary cell geometry to yield 68.4% below 100 ms. Similarly, Knot 9 ($x = -0.308$, $P_0 = 492.4$ ms) straddles 0.5 s; discrete summation over Knots 9–13 includes the full discrete cell mass of Knot 9 (0.058), yielding 19.3% (or 13.5% if Knot 9 is excluded), while continuous integration splits the cell to yield 10.8%–11.1%. On the continuous 240 × 240 PDE grid (`DemographicsEngine`), 2D multilinear interpolation followed by continuum Gaussian filtering ($\sigma = 1.5$ grid cells) reproduces these continuous metrics: median $P_0 = 58.9$ ms, peak at 47.9 ms, interquartile range [45.5, 162.4] ms, 90th percentile $P_{0,90\%} = 0.54$ s, with 10.8% at $P_0 \geq 0.50$ s (11.1% above Knot 9), 3.5% at $P_0 \geq 1.0$ s (3.8% discrete), and 68.4% below 100 ms (60.8% Knots 0–3). The baseline empirical reconstruction thus possesses an intrinsic tail with 10.8%–11.1% born above 0.5 s. Crucially, the limit $f_{\rm slow}^{\rm birth} \leq 22\%$ established in Bindra (2026b) defines the 95% upper bound on the *total* slow-birth fraction; the allowable headroom for an incremental discrete injection is at most $\Delta f_{\rm slow} \leq 22\% - 10.8\% \approx 11.2\%$ (or 10.9% relative to Knot 9). The counterfactual scenarios evaluated in Section 4.4 test adding incremental discrete channels of +20% and +50% (raising the total slow fraction to ≈ 31% and ≈ 61%), which violate this ceiling and induce severe demographic degradation.

We transparently disclose that the discrete inversion concentrates probability mass on two boundary knots: 18.1% of birth mass sits on the lower period knot at $P_0 = 35.5$ ms ($x = -1.450$), and 30.4% sits on the upper magnetic field knot at $B_0 = 10^{13.40}$ G $= 2.5 \times 10^{13}$ G ($y = 13.400$). Because the inversion grid was bounded at $B_0 \leq 10^{13.40}$ G, any births occurring at higher dipole fields are absorbed into this upper boundary knot. If an extended tail of pulsars is born with magnetar-strength dipole fields ($B_0 \gtrsim 3$–5×10^{13} G), their intensely decelerating torques ($\dot{P} \propto B^2/P$) would accelerate their transit across the P–$\dot{P}$ diagram, resulting in individual transit times $t_{\rm death} \sim 1$–2 Myr and reducing the population-averaged active lifetime toward 5.33–6.16 Myr ($N_{\rm Gal}^{\rm active} = (0.60$–$0.69) \times 10^5 \times (f_{\rm radio}/0.75)$), pushing toward the lower end of our overall systematic parameter envelope ($[0.49, 1.25] \times 10^5$). Conversely, Crab-like births below 35 ms traverse the sub-35 ms regime in $< 10^4$ yr, so bounding the knot lattice at 35.5 ms introduces negligible distortion to the second-scale equilibrium bulk. In phase space, the characteristic transport parameters are: crustal field decay rate $h_0 = 0.0150$ Myr^{-1} ($\tau_B \approx 67$ Myr), global secular escape rate $\kappa = 0.0080$ Myr^{-1} ($\tau_\kappa = 125$ Myr, modeling unbinding and gradual evaporation into the distant halo), and crustal bottom floor $\log_{10}(B_{\rm floor}/{\rm G}) = 11.20$ ($B_{\rm floor} \approx 1.6 \times 10^{11}$ G). The full 14 × 15 matrix $S_{ij} \equiv S(x_i, y_j)$ is available as a machine-readable data file in the accompanying repository.

DATA AND CODE AVAILABILITY

The complete Python continuum analysis pipeline, Fortran 90/2008 continuous-time simulation code, pre-computed steady-state snapshots, and figure generation scripts are openly available on GitHub at https://github.com/anshveerbindra/Pulsar-Demographics-Continuity. The observational pulsar data utilized in this study are derived from the Australia Telescope National Facility (ATNF) Pulsar Catalogue version 2.6.4 (https://www.atnf.csiro.au/research/pulsar/psrcat/).

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Table 4. Marginal Birth Injection Distributions across Knot Locations. Unweighted discrete column/row array sums are shown; continuous 2D trapezoidal integration of the continuous injection rate density $S(x, y)$ accounts for boundary cell quadrature weights ($\frac{1}{2}\Delta x$), yielding cumulative birth fractions of 68.4% below 100 ms, 10.8%–11.1% above 0.5 s, and 3.5%–3.6% above 1.0 s (3.8% discrete).

Spin Period Knots		**Magnetic Field Knots**	
$\log_{10}(P_0/\mathrm{s})$	Marginal $S(x)$	$\log_{10}(B_0/\mathrm{G})$	Marginal $S(y)$
−1.450	0.181	10.650	0.000
−1.323	0.250	10.846	0.000
−1.196	0.155	11.043	0.000
−1.069	0.022	11.239	0.000
−0.942	0.046	11.436	0.002
−0.815	0.037	11.632	0.005
−0.688	0.030	11.829	0.011
−0.562	0.036	12.025	0.038
−0.435	0.049	12.221	0.053
−0.308	0.058	12.418	0.117
−0.181	0.043	12.614	0.140
−0.054	0.035	12.811	0.082
+0.073	0.034	13.007	0.185
+0.200	0.023	13.204	0.063
—	—	13.400	0.304

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